Activation-Weighted Low-Rank Compression: Likelihood, Drift, and Practical Cost

PREPRINT REVIEW DRAFT -- no venue assumed

No established global-preservation or practical-compression claim. Original FP16 readiness remains failed. Native-tokenizer perplexities are compared within models only. Partial and cancelled attempts remain evidence, not completed cohorts.

Equations use readable ASCII notation (beta, rho, lambda; dagger denotes the pseudoinverse). The accompanying standalone manuscript.tex provides exact typeset mathematical source. This PDF is independently rendered from the same Markdown and measured figures.

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Abstract

This report examines whether matched activation-weighted low-rank approximations preserve global transformer behavior. It retains weighting/anchoring controls, full-stack and composition measurements, separate BF16/FP8 numerical controls, and the user-amended hybrid-architecture panel. Tables below are generated from retained artifacts; failed/incomplete attempts remain visible. Targeted weight-count reduction is separated from likelihood, representation drift and measured execution costs. The evidence does not establish reliable global preservation or a consistent anchoring advantage over matched whitening. Promotion remains disabled; a terminal dispatcher does not imply every scientific cohort completed.

Study status and scope

Dispatcher: completed with incomplete cohorts. Report mode: terminal orchestration report.

The amended panel uses Gemma, Qwen3.5-0.8B and LFM2.5-1.2B-Instruct; GPT-2 is excluded. Only full-attention output projections are targeted in the hybrids. Native-tokenizer PPL is compared within each model. Validation frontier ranks and primary test ranks remain separate; no selection is made using confirmation outcomes.

Later BF16-only practical amendment

After stopping the historical FP16/INT8 practical stage, the user authorized only BF16 model testing with at most two seeds. The new practical cohort has five declared conditions: a BF16 dense teacher, and BF16 whitening/anchor models for seeds 11 and 23. Native BF16 measured-output factor preparation retains the frozen objective and rank. Historical other-format and three-seed evidence is preserved, not relabeled or pooled with this cohort. Higher-precision fitting/metric accumulation is numerical computation, not another model-format test. Two seeds cannot satisfy the original three-seed promotion gate. The stopped stage remains charged against the original shared practical cap.

Declared numerical-control omissions

The continuation formally omits unfinished dense_fp8_e4m3 and dense_fp8_e5m2 controls. All seven saved controls, including failed gates, and the cancelled attempt remain preserved. This scope closure does not claim that the original nine-control cohort completed or that native FP8 model inference was tested.

Mathematical methods and frozen constants

Let W in R^{o x i} and X in R^{i x n}. Write A=UV with rank at most r. The anchored objective is ||(W-A)X||_F^2+beta n||W-A||_F^2.

The replacement is the attention-output projection after head outputs are combined; attention heads themselves are not pruned. Fitting is offline, and inference applies V followed by U in place of W. Anchoring adds no runtime branch or extra factor parameters: it changes only the fitting geometry.

For each Gemma target, the dense matrix has 1152 x 1024 = 1,179,648 weights. Rank 384 stores 1152 x 384 + 384 x 1024 = 835,584 weights: 344,064 fewer, or 29.17% of that projection. Across 26 targets the reduction is 8,945,664 weights. These are targeted weight counts, not a 29.17% whole-model reduction; embeddings, MLPs and other untargeted weights remain.

Set H=XX^T+beta nI and C=WH for exact teacher outputs. For measured outputs subtract the teacher bias first: Y_b=Y-b1^T, then C=Y_bX^T+beta nW. The extra-ridge solver adds lambda I to the geometry, with lambda=rho mean(eig(H)). For G=H+lambda I, form A_full=CG^dagger and truncate A_fullG^{1/2} to rank r; multiply back by G^{dagger/2}. Thus ridge-free full-rank anchored recovery gives W on retained spectral directions, while positive extra ridge generally shrinks it.

The implementation computes weighted covariance, eigendecomposition and weighted truncation in FP64, then returns FP32 factors for the declared storage cast. It clamps negative numerical eigenvalues to zero and retains regularized eigenvalues strictly above rcond lambda_max(G). Plain SVD uses FP32. Fixed-rank diagonal ASVD uses s_j=max(mean_t|X_{jt}|,10^{-8})^{0.5}, truncates Wdiag(s), and scales back. Factors use the recorded unbalanced convention unless a run contract explicitly declares a separate convention.

The predeclared validation anchor sweep is beta={0,0.03,0.1,0.3,1}; the extra-ridge fractions are rho={0,0.001,0.01} at beta=0.1. Validation NLL ties within 1e-6 prefer less regularization. Native evaluation uses context/stride 512/256, full shifted targets including the tail, and 16 disjoint 512-token calibration chunks for seeds 11/23/37. The primary Gemma attention-output rank is 384. The amended hybrid rank is floor((17/24)*out*in/(out+in)); frontier ranks are floor(min(out,in)/divisor) for validation-only divisors2/3/4. Exact observed run constants and factor precisions follow below; missing fields remain unrecorded.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 2; match by Row ID.

Row ID	Run	Precision	Main rank	Pseudoinverse rcond
R001	architecture-replication-20261014-082807-5c73f669	bfloat16	shape-derived by 17/24 factor/dense fraction	1e-07
R002	architecture-replication-20261014-093845-97378b5e	bfloat16	shape-derived by 17/24 factor/dense fraction	1e-07
R003	architecture-replication-20261014-103759-8d32ab84	bfloat16	shape-derived by 17/24 factor/dense fraction	1e-07
R004	baselines-20261013-175255-6a00d4e8	float16	384	1e-07
R005	composition-20261013-185512-2bffd477	float16	384	1e-07
R006	composition-20261014-032120-486d799f	float16	384	1e-07
R007	composition-20261014-054423-dbd821ff	float16	384	1e-07
R008	practical-20261014-113642-0fe078b5	float16	384	1e-07
R009	practical-bf16-20261014-135107-82168ad5	bfloat16	384	1e-07
R010	precision-readiness-20261014-060840-6db37daf	float16	384	1e-07
R011	regularization-20261013-171941-804f4c2e	float16	384	1e-07

Panel 2 of 2; match by Row ID.

Row ID	Run	Config beta	Config ridge fraction	Selected solver / baseline
R001	architecture-replication-20261014-082807-5c73f669	unmeasured	unmeasured	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}}
R002	architecture-replication-20261014-093845-97378b5e	unmeasured	unmeasured	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}}
R003	architecture-replication-20261014-103759-8d32ab84	unmeasured	unmeasured	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}}
R004	baselines-20261013-175255-6a00d4e8	0.1	0.01	{"baseline": "SVD and fixed-rank ASVD / whitening-only components, same FP16/rank", "solver": null}

Row ID	Run	Config beta	Config ridge fraction	Selected solver / baseline
R005	composition-20261013-185512-2bffd477	0.1	0.01	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null}
R006	composition-20261014-032120-486d799f	0.1	0.01	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null}
R007	composition-20261014-054423-dbd821ff	0.1	0.01	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null}
R008	practical-20261014-113642-0fe078b5	0.1	0.01	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": null}
R009	practical-bf16-20261014-135107-82168ad5	0.03	0	{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "solver": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}}
R010	precision-readiness-20261014-060840-6db37daf	0.1	0.01	{"baseline": "untouched dense BF16 teacher and copied BF16 control", "solver": null}
R011	regularization-20261013-171941-804f4c2e	0.1	0.01	{"baseline": "SVD and fixed-rank ASVD / whitening-only components, same FP16/rank", "solver": null}

Related work and scope of the planned evidence

Data-aware low-rank compression predates this study. DRONE (https://proceedings.neurips.cc/paper/2021/hash/f56de5ef149cf0aedcc8f4797031e229-Abstract.html) derives a closed-form approximation informed by layer representations in BERT [@chen2021drone]. Fisher-weighted factorization (https://arxiv.org/abs/2207.00112) uses parameter importance to address disagreement between matrix reconstruction and task accuracy [@hsu2022weighted]. These are context for the present question: low local weight error alone cannot establish preserved global computation.

ASVD (https://arxiv.org/abs/2312.05821) adapts decomposition to activation distributions and layer sensitivity [@yuan2023asvd].

SVD-LLM (https://arxiv.org/abs/2403.07378) combines truncation-aware whitening with sequential updates after compression [@wang2024svdllm]. Our fixed-rank diagonal ASVD and whitening-only SVD-LLM isolate their weighting components under a matched factor budget. They are not full-pipeline reproductions.

SVD-LLM V2 (https://aclanthology.org/2025.naacl-long.217/) changes rank allocation and singular-value truncation [@wang2025svdllmv2], so it belongs in a later adaptive-budget study rather than a claim that fixed per-matrix ranks reproduce its complete procedure.

Global composition is already an explicit target of related work.

SAES-SVD (https://arxiv.org/abs/2602.03051) addresses accumulated and local errors [@hu2026saessvd]; LACE-SVD (https://arxiv.org/abs/2607.03057) combines loss-aware allocation with residual-stream correction [@liu2026lacesvd].

Our smaller-group and downstream-drift comparisons test the location and extent

of composition damage, without claiming to resolve it. The next comparator is SAES-SVD if cumulative drift remains the decisive failure.

Quantization Error Propagation (https://arxiv.org/abs/2504.09629) demonstrates why accumulated errors also matter in layer-wise quantization [@arai2025qep]. That connection motivates measuring downstream behavior, but does not equate low-rank truncation with quantization or transfer an existing method's results.

Loss-aware global allocation provides further distinctions from this study's fixed-rank setup. Zero Sum SVD (https://arxiv.org/abs/2602.02848) uses whitened component sensitivities under a global pruning rule [@abbasi2026zerosum]. IO-SVD (https://arxiv.org/abs/2605.15626) adds an output-side predictive metric and heterogeneous rank selection [@abbasi2026iosvd]. These motivate future equal-total-budget comparisons. They do not establish novelty for an anchored input-covariance truncation or for the current extra-ridge solver.

The practical comparison distinguishes storage, accuracy, and execution cost. GPTQ (https://arxiv.org/abs/2210.17323) uses approximate second-order information for post-training quantization [@frantar2022gptq].

AWQ (https://arxiv.org/abs/2306.00978) uses activation-informed channel protection and an inference implementation [@lin2023awq].

SparseGPT (https://arxiv.org/abs/2301.00774) studies one-shot pruning [@frantar2023sparsegpt]. These are different representations, not matched fixed-rank baselines in the first wave. Their practical emphasis motivates recording actual checkpoint bytes and warmed end-to-end timings.

EoRA (https://arxiv.org/abs/2410.21271) compensates compressed models with eigenspace-informed low-rank branches [@liu2024eora].

SVDQuant (https://arxiv.org/abs/2411.05007) absorbs outliers in a high-precision low-rank branch while quantizing a residual branch, and identifies the runtime overhead of naive separate branch execution [@li2024svdquant]. SVDQuant's published scope is diffusion models. It supplies a system-design warning for future hybrid experiments here, not evidence of LLM quality or speedups.

CKA (https://proceedings.mlr.press/v97/kornblith19a.html) measures alignment between representation similarities [@kornblith2019cka]. We use it as a secondary diagnostic beside the predeclared normalized residual squared error; neither representation similarity nor norm preservation substitutes for held-out NLL or logit KL. Temperature scaling (https://proceedings.mlr.press/v70/guo17a.html) is a validation-fitted calibration control [@guo2017calibration], adapted here to next-token probabilities. Improvement after temperature fitting would not by itself identify an LLM denoising mechanism.

Evaluation conventions can change apparent model rankings; the reproducibility guidance in Biderman et al. (https://arxiv.org/abs/2405.14782) motivates pinned assets, task formats and declared target masking [@biderman2024evaluation].

Pointer Sentinel Mixture Models (https://arxiv.org/abs/1609.07843) introduced

WikiText [@merity2016wikitext]. Because WikiText has already informed this project, its present test evaluation is regression evidence. The later fresh confirmation samples a locked C4 shard, whose provenance is the T5 study (https://jmlr.org/papers/v21/20-074.html) [@raffel2020t5]. Neither its sampled confirmation nor the initial WikiText wave represents all text domains. HellaSwag (https://aclanthology.org/P19-1472/) supplies a commonsense completion benchmark [@zellers2019hellaswag], and ARC (https://arxiv.org/abs/1803.05457) supplies science-question reasoning with an explicit Challenge subset [@clark2018arc]. Those downstream evaluations remain pending; no downstream accuracy claim follows from the readiness controls.

Recorded numerical controls

FP16 SVD readiness remains failed. Later BF16/FP8 variants are separate numerical cohorts, not repaired FP16 evidence. FP8 codes are decoded before BF16 multiplication; native tiny-GEMM support does not establish native model speed.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 2; match by Row ID.

Row ID	Run / state	Variant	NLL	PPL
R001	precision-readiness-20261014-060840-6db37daf / cancelled	teacher_bf16	3.4515495	31.5492399
R002	precision-readiness-20261014-060840-6db37daf / cancelled	copy_bf16	3.4515495	31.5492399
R003	precision-readiness-20261014-060840-6db37daf / cancelled	identity_bf16	3.4515495	31.5492399
R004	precision-readiness-20261014-060840-6db37daf / cancelled	svd_bf16	3.45265812	31.5842357
R005	precision-readiness-20261014-060840-6db37daf / cancelled	svd_fp32_compute	3.45158372	31.5503197
R006	precision-readiness-20261014-060840-6db37daf / cancelled	svd_fp8_e4m3	3.45542964	31.6718934
R007	precision-readiness-20261014-060840-6db37daf / cancelled	svd_fp8_e5m2	3.47055948	32.1547273
R008	readiness-20261013-164122-cdc919e9 / failed	teacher	3.45156254	31.5496515
R009	readiness-20261013-164122-cdc919e9 / failed	copied_teacher	3.45156254	31.5496515
R010	readiness-20261013-164407-bda5b4d9 / completed	teacher	3.45156254	31.5496515
R011	readiness-20261013-164407-bda5b4d9 / completed	copied_teacher	3.45156254	31.5496515
R012	readiness-20261013-164407-bda5b4d9 / completed	full_rank_factorized	3.45156254	31.5496515
R013	svd-diagnosis-20261013-165146-c3671715 / completed	teacher	3.45156254	31.5496515
R014	svd-diagnosis-20261013-165146-c3671715 / completed	dense_svd_fp32	3.4516607	31.5527485
R015	svd-diagnosis-20261013-165146-c3671715 / completed	dense_svd_fp16	3.45170481	31.5541402
R016	svd-diagnosis-20261013-165146-c3671715 / completed	unbalanced_fp16	3.45186016	31.5590426
R017	svd-diagnosis-20261013-165146-c3671715 / completed	balanced_fp16	3.45184835	31.5586697
R018	svd-diagnosis-20261013-165146-c3671715 / completed	float32_execution	3.4517713	31.5562382

Panel 2 of 2; match by Row ID.

Row ID	Run / state	Maximum probe error	Probe gate	NLL gate
R001	precision-readiness-20261014-060840-6db37daf / cancelled	0	True	True
R002	precision-readiness-20261014-060840-6db37daf / cancelled	0	True	True
R003	precision-readiness-20261014-060840-6db37daf / cancelled	0	True	True
R004	precision-readiness-20261014-060840-6db37daf / cancelled	1.0625	False	False
R005	precision-readiness-20261014-060840-6db37daf / cancelled	1.15625	False	True
R006	precision-readiness-20261014-060840-6db37daf / cancelled	9.73046875	False	False
R007	precision-readiness-20261014-060840-6db37daf / cancelled	7.8671875	False	False
R008	readiness-20261013-164122-cdc919e9 / failed	0	True	unrecorded
R009	readiness-20261013-164122-cdc919e9 / failed	0	True	unrecorded
R010	readiness-20261013-164407-bda5b4d9 / completed	0	True	unrecorded
R011	readiness-20261013-164407-bda5b4d9 / completed	0	True	unrecorded
R012	readiness-20261013-164407-bda5b4d9 / completed	0	True	unrecorded
R013	svd-diagnosis-20261013-165146-c3671715 / completed	unmeasured	unrecorded	unrecorded
R014	svd-diagnosis-20261013-165146-c3671715 / completed	0.1171875	True	True
R015	svd-diagnosis-20261013-165146-c3671715 / completed	0.140625	False	True
R016	svd-diagnosis-20261013-165146-c3671715 / completed	0.15234375	False	True
R017	svd-diagnosis-20261013-165146-c3671715 / completed	0.181640625	False	True
R018	svd-diagnosis-20261013-165146-c3671715 / completed	0.1171875	True	True

Validation regularization controls

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R001	regularization-20261013-171941-804f4c2e / completed	{"method": "svd"}	{}	[11]
R002	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "fullrank_ridge", "ridge_fraction": 0.01}	{}	[11]
R003	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{}	[11]
R004	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{}	[11]

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R005	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "anchored", "ridge_fraction": 0.0}	{}	[11]
R006	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.3, "method": "anchored", "ridge_fraction": 0.0}	{}	[11]
R007	regularization-20261013-171941-804f4c2e / completed	{"beta": 1.0, "method": "anchored", "ridge_fraction": 0.0}	{}	[11]
R008	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.0}	{}	[11]
R009	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.001}	{}	[11]
R010	regularization-20261013-171941-804f4c2e / completed	{"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.01}	{}	[11]
R011	regularization-20261013-171941-804f4c2e / completed	{"method": "asvd"}	{}	[11]

Panel 2 of 3; match by Row ID.

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R001	regularization-20261013-171941-804f4c2e / completed	5.14486078	unmeasured	unmeasured
R002	regularization-20261013-171941-804f4c2e / completed	3.49923448	unmeasured	unmeasured
R003	regularization-20261013-171941-804f4c2e / completed	3.60078899	unmeasured	unmeasured
R004	regularization-20261013-171941-804f4c2e / completed	3.59662239	unmeasured	unmeasured
R005	regularization-20261013-171941-804f4c2e / completed	3.59675176	unmeasured	unmeasured
R006	regularization-20261013-171941-804f4c2e / completed	3.607465	unmeasured	unmeasured
R007	regularization-20261013-171941-804f4c2e / completed	3.61579886	unmeasured	unmeasured
R008	regularization-20261013-171941-804f4c2e / completed	3.59685438	unmeasured	unmeasured
R009	regularization-20261013-171941-804f4c2e / completed	3.59727845	unmeasured	unmeasured
R010	regularization-20261013-171941-804f4c2e / completed	3.60682971	unmeasured	unmeasured
R011	regularization-20261013-171941-804f4c2e / completed	3.73607467	unmeasured	unmeasured

Panel 3 of 3; match by Row ID.

Row ID	Run / state	Drift seed SD
R001	regularization-20261013-171941-804f4c2e / completed	unmeasured
R002	regularization-20261013-171941-804f4c2e / completed	unmeasured
R003	regularization-20261013-171941-804f4c2e / completed	unmeasured
R004	regularization-20261013-171941-804f4c2e / completed	unmeasured
R005	regularization-20261013-171941-804f4c2e / completed	unmeasured
R006	regularization-20261013-171941-804f4c2e / completed	unmeasured
R007	regularization-20261013-171941-804f4c2e / completed	unmeasured
R008	regularization-20261013-171941-804f4c2e / completed	unmeasured

Row ID	Run / state	Drift seed SD
R009	regularization-20261013-171941-804f4c2e / completed	unmeasured
R010	regularization-20261013-171941-804f4c2e / completed	unmeasured
R011	regularization-20261013-171941-804f4c2e / completed	unmeasured

Gemma full-stack component comparisons

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R001	baselines-20261013-175255-6a00d4e8 / completed	{"method": "svd"}	{}	[11]
R002	baselines-20261013-175255-6a00d4e8 / completed	{"method": "asvd"}	{}	[11, 23, 37]
R003	baselines-20261013-175255-6a00d4e8 / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{}	[11, 23, 37]
R004	baselines-20261013-175255-6a00d4e8 / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{}	[11, 23, 37]
R005	baselines-20261013-175255-6a00d4e8 / completed	{"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.01}	{}	[11, 23, 37]

Panel 2 of 3; match by Row ID.

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R001	baselines-20261013-175255-6a00d4e8 / completed	5.08868112	unmeasured	0.0742475831
R002	baselines-20261013-175255-6a00d4e8 / completed	3.69446606	0.00241956806	0.0309717453
R003	baselines-20261013-175255-6a00d4e8 / completed	3.57281383	0.0118191598	0.0106375166
R004	baselines-20261013-175255-6a00d4e8 / completed	3.56592113	0.00957592299	0.0105470011
R005	baselines-20261013-175255-6a00d4e8 / completed	3.57903427	0.00852077408	0.0118359709

Panel 3 of 3; match by Row ID.

Row ID	Run / state	Drift seed SD
R001	baselines-20261013-175255-6a00d4e8 / completed	unmeasured
R002	baselines-20261013-175255-6a00d4e8 / completed	0.00019731531
R003	baselines-20261013-175255-6a00d4e8 / completed	0.000382136395
R004	baselines-20261013-175255-6a00d4e8 / completed	0.000492296849
R005	baselines-20261013-175255-6a00d4e8 / completed	0.000138552104

Composition and group heterogeneity

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R001	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R002	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R003	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R004	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R005	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R006	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R007	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R008	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R009	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R010	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R011	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R012	composition-20261013-185512-2bffd477 / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11]
R013	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R014	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R015	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R016	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R017	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R018	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R019	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R020	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R021	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R022	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R023	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R024	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R025	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3], "rank": 384, "split": "test"}	[11, 23, 37]
R026	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3], "rank": 384, "split": "test"}	[11, 23, 37]
R027	composition-20261014-032120-486d799f / interrupted	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [11, 12, 13, 14], "rank": 384, "split": "test"}	[11, 23, 37]
R028	composition-20261014-032120-486d799f / interrupted	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [11, 12, 13, 14], "rank": 384, "split": "test"}	[11, 23]
R029	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R030	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0], "rank": 384, "split": "test"}	[11, 23, 37]
R031	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R032	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12], "rank": 384, "split": "test"}	[11, 23, 37]
R033	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R034	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [25], "rank": 384, "split": "test"}	[11, 23, 37]
R035	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R036	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1], "rank": 384, "split": "test"}	[11, 23, 37]
R037	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R038	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [12, 13], "rank": 384, "split": "test"}	[11, 23, 37]
R039	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R040	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R041	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3], "rank": 384, "split": "test"}	[11, 23, 37]
R042	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3], "rank": 384, "split": "test"}	[11, 23, 37]

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R043	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [11, 12, 13, 14], "rank": 384, "split": "test"}	[11, 23, 37]
R044	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [11, 12, 13, 14], "rank": 384, "split": "test"}	[11, 23, 37]
R045	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [22, 23, 24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R046	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [22, 23, 24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R047	composition-20261014-054423-dbd821ff / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25], "rank": 384, "split": "test"}	[11, 23, 37]
R048	composition-20261014-054423-dbd821ff / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"group": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25], "rank": 384, "split": "test"}	[11, 23, 37]

Panel 2 of 3; match by Row ID.

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R001	composition-20261013-185512-2bffd477 / interrupted	3.45206785	0.000410778252	0.00207792164
R002	composition-20261013-185512-2bffd477 / interrupted	3.45201942	0.000409256997	0.00207452887
R003	composition-20261013-185512-2bffd477 / interrupted	3.45825421	0.00132397251	8.76745161e-05
R004	composition-20261013-185512-2bffd477 / interrupted	3.4579111	0.00143757277	8.91802058e-05
R005	composition-20261013-185512-2bffd477 / interrupted	3.45142891	0.000329325004	1.69831209e-06
R006	composition-20261013-185512-2bffd477 / interrupted	3.45122641	0.000131137763	2.023157e-06
R007	composition-20261013-185512-2bffd477 / interrupted	3.45442908	0.00205462111	0.00365644585
R008	composition-20261013-185512-2bffd477 / interrupted	3.45438131	0.00214728679	0.00356760921
R009	composition-20261013-185512-2bffd477 / interrupted	3.46353003	0.00115146509	0.000141608504
R010	composition-20261013-185512-2bffd477 / interrupted	3.46353582	0.00129669463	0.000144554107
R011	composition-20261013-185512-2bffd477 / interrupted	3.45212529	0.000451146203	1.23266207e-05
R012	composition-20261013-185512-2bffd477 / interrupted	3.45195513	unmeasured	1.21042136e-05
R013	composition-20261014-032120-486d799f / interrupted	3.45206785	0.000410778252	0.00207792164
R014	composition-20261014-032120-486d799f / interrupted	3.45201942	0.000409256997	0.00207452887
R015	composition-20261014-032120-486d799f / interrupted	3.45825421	0.00132397251	8.76745161e-05
R016	composition-20261014-032120-486d799f / interrupted	3.4579111	0.00143757277	8.91802058e-05
R017	composition-20261014-032120-486d799f / interrupted	3.45142891	0.000329325004	1.69831209e-06
R018	composition-20261014-032120-486d799f / interrupted	3.45122641	0.000131137763	2.023157e-06

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R019	composition-20261014-032120-486d799f / interrupted	3.45442908	0.00205462111	0.00365644585
R020	composition-20261014-032120-486d799f / interrupted	3.45438131	0.00214728679	0.00356760921
R021	composition-20261014-032120-486d799f / interrupted	3.46353003	0.00115146509	0.000141608504
R022	composition-20261014-032120-486d799f / interrupted	3.46353582	0.00129669463	0.000144554107
R023	composition-20261014-032120-486d799f / interrupted	3.45212529	0.000451146203	1.23266207e-05
R024	composition-20261014-032120-486d799f / interrupted	3.45194198	0.000101102007	1.22970593e-05
R025	composition-20261014-032120-486d799f / interrupted	3.46224947	0.000734496385	0.0061389891
R026	composition-20261014-032120-486d799f / interrupted	3.4619553	0.00102918121	0.00600512161
R027	composition-20261014-032120-486d799f / interrupted	3.47310172	0.000470679639	0.00038058521
R028	composition-20261014-032120-486d799f / interrupted	3.4765793	1.58332576e-05	0.000438405635
R029	composition-20261014-054423-dbd821ff / completed	3.45206785	0.000410778252	0.00207792164
R030	composition-20261014-054423-dbd821ff / completed	3.45201942	0.000409256997	0.00207452887
R031	composition-20261014-054423-dbd821ff / completed	3.45825421	0.00132397251	8.76745161e-05
R032	composition-20261014-054423-dbd821ff / completed	3.4579111	0.00143757277	8.91802058e-05
R033	composition-20261014-054423-dbd821ff / completed	3.45142891	0.000329325004	1.69831209e-06
R034	composition-20261014-054423-dbd821ff / completed	3.45122641	0.000131137763	2.023157e-06
R035	composition-20261014-054423-dbd821ff / completed	3.45442908	0.00205462111	0.00365644585
R036	composition-20261014-054423-dbd821ff / completed	3.45438131	0.00214728679	0.00356760921
R037	composition-20261014-054423-dbd821ff / completed	3.46353003	0.00115146509	0.000141608504
R038	composition-20261014-054423-dbd821ff / completed	3.46353582	0.00129669463	0.000144554107
R039	composition-20261014-054423-dbd821ff / completed	3.45212529	0.000451146203	1.23266207e-05
R040	composition-20261014-054423-dbd821ff / completed	3.45194198	0.000101102007	1.22970593e-05
R041	composition-20261014-054423-dbd821ff / completed	3.46224947	0.000734496385	0.0061389891
R042	composition-20261014-054423-dbd821ff / completed	3.4619553	0.00102918121	0.00600512161
R043	composition-20261014-054423-dbd821ff / completed	3.47310172	0.000470679639	0.00038058521
R044	composition-20261014-054423-dbd821ff / completed	3.47582544	0.00130576482	0.000449140822
R045	composition-20261014-054423-dbd821ff / completed	3.45396348	0.00136122732	0.000123680895
R046	composition-20261014-054423-dbd821ff / completed	3.4529696	0.000385430754	0.00011315281
R047	composition-20261014-054423-dbd821ff / completed	3.57281383	0.0118191598	0.0106375166

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R048	composition-20261014-054423-dbd821ff / completed	3.56592113	0.00957592299	0.0105470011

Panel 3 of 3; match by Row ID.

Row ID	Run / state	Drift seed SD
R001	composition-20261013-185512-2bffd477 / interrupted	0.00017472334
R002	composition-20261013-185512-2bffd477 / interrupted	0.000178507413
R003	composition-20261013-185512-2bffd477 / interrupted	2.4463997e-06
R004	composition-20261013-185512-2bffd477 / interrupted	2.24916162e-06
R005	composition-20261013-185512-2bffd477 / interrupted	9.05406025e-08
R006	composition-20261013-185512-2bffd477 / interrupted	9.42299447e-08
R007	composition-20261013-185512-2bffd477 / interrupted	0.000137627131
R008	composition-20261013-185512-2bffd477 / interrupted	0.000238304358
R009	composition-20261013-185512-2bffd477 / interrupted	4.06719991e-06
R010	composition-20261013-185512-2bffd477 / interrupted	4.09979657e-06
R011	composition-20261013-185512-2bffd477 / interrupted	3.68456941e-07
R012	composition-20261013-185512-2bffd477 / interrupted	unmeasured
R013	composition-20261014-032120-486d799f / interrupted	0.00017472334
R014	composition-20261014-032120-486d799f / interrupted	0.000178507413
R015	composition-20261014-032120-486d799f / interrupted	2.4463997e-06
R016	composition-20261014-032120-486d799f / interrupted	2.24916162e-06
R017	composition-20261014-032120-486d799f / interrupted	9.05406025e-08
R018	composition-20261014-032120-486d799f / interrupted	9.42299447e-08
R019	composition-20261014-032120-486d799f / interrupted	0.000137627131
R020	composition-20261014-032120-486d799f / interrupted	0.000238304358
R021	composition-20261014-032120-486d799f / interrupted	4.06719991e-06
R022	composition-20261014-032120-486d799f / interrupted	4.09979657e-06
R023	composition-20261014-032120-486d799f / interrupted	3.68456941e-07
R024	composition-20261014-032120-486d799f / interrupted	3.91576035e-07
R025	composition-20261014-032120-486d799f / interrupted	0.000142663396
R026	composition-20261014-032120-486d799f / interrupted	0.000161076046
R027	composition-20261014-032120-486d799f / interrupted	1.72440575e-05
R028	composition-20261014-032120-486d799f / interrupted	3.31525237e-07
R029	composition-20261014-054423-dbd821ff / completed	0.00017472334
R030	composition-20261014-054423-dbd821ff / completed	0.000178507413
R031	composition-20261014-054423-dbd821ff / completed	2.4463997e-06
R032	composition-20261014-054423-dbd821ff / completed	2.24916162e-06
R033	composition-20261014-054423-dbd821ff / completed	9.05406025e-08
R034	composition-20261014-054423-dbd821ff / completed	9.42299447e-08
R035	composition-20261014-054423-dbd821ff / completed	0.000137627131
R036	composition-20261014-054423-dbd821ff / completed	0.000238304358
R037	composition-20261014-054423-dbd821ff / completed	4.06719991e-06
R038	composition-20261014-054423-dbd821ff / completed	4.09979657e-06
R039	composition-20261014-054423-dbd821ff / completed	3.68456941e-07

Row ID	Run / state	Drift seed SD
R040	composition-20261014-054423-dbd821ff / completed	3.91576035e-07
R041	composition-20261014-054423-dbd821ff / completed	0.000142663396
R042	composition-20261014-054423-dbd821ff / completed	0.000161076046
R043	composition-20261014-054423-dbd821ff / completed	1.72440575e-05
R044	composition-20261014-054423-dbd821ff / completed	1.85953677e-05
R045	composition-20261014-054423-dbd821ff / completed	1.21816693e-06
R046	composition-20261014-054423-dbd821ff / completed	2.34008283e-06
R047	composition-20261014-054423-dbd821ff / completed	0.000382136395
R048	composition-20261014-054423-dbd821ff / completed	0.000492296849

Group-dependent outcomes must not be pooled into a universal-improvement claim. A lower full-stack NLL with unchanged/worse residual drift does not satisfy the predeclared joint gate. Three paired seeds with both higher NLL and higher drift weaken the candidate only at that measured scope.

Amended hybrid architecture and rank-frontier comparisons

Every row identifies its attempt and scope. Mean ± sample standard deviation is across available calibration seeds; a single seed has no estimated seed standard deviation. Incomplete attempts and partial seed coverage do not constitute a full cohort.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 3; match by Row ID.

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R001	architecture-replication-20261014-093845-97378b5e / incomplete	{"method": "teacher"}	{"rank": null, "rank_divisor": null, "recipe_ranks": [null, null, null, null, null, null], "split": "test"}	[11]
R002	architecture-replication-20261014-093845-97378b5e / incomplete	{"method": "svd"}	{"rank": 483, "rank_divisor": null, "recipe_ranks": [483, 483, 483, 483, 483, 483], "split": "test"}	[11]
R003	architecture-replication-20261014-093845-97378b5e / incomplete	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"rank": 483, "rank_divisor": null, "recipe_ranks": [483, 483, 483, 483, 483, 483], "split": "test"}	[11, 23, 37]
R004	architecture-replication-20261014-093845-97378b5e / incomplete	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"rank": 483, "rank_divisor": null, "recipe_ranks": [483, 483, 483, 483, 483, 483], "split": "test"}	[11, 23, 37]
R005	architecture-replication-20261014-093845-97378b5e / incomplete	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"rank": 512, "rank_divisor": 2, "recipe_ranks": [512, 512, 512, 512, 512, 512], "split": "validation"}	[11, 23, 37]
R006	architecture-replication-20261014-093845-97378b5e / incomplete	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"rank": 512, "rank_divisor": 2, "recipe_ranks": [512, 512, 512, 512, 512, 512], "split": "validation"}	[11, 23, 37]
R007	architecture-replication-20261014-093845-97378b5e / incomplete	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"rank": 341, "rank_divisor": 3, "recipe_ranks": [341, 341, 341, 341, 341, 341], "split": "validation"}	[11, 23]
R008	architecture-replication-20261014-093845-97378b5e / incomplete	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"rank": 341, "rank_divisor": 3, "recipe_ranks": [341, 341, 341, 341, 341, 341], "split": "validation"}	[11, 23]
R009	architecture-replication-20261014-103759-8d32ab84 / completed	{"method": "teacher"}	{"rank": null, "rank_divisor": null, "recipe_ranks": [null, null, null, null, null, null], "split": "test"}	[11]
R010	architecture-replication-20261014-103759-8d32ab84 / completed	{"method": "svd"}	{"rank": 725, "rank_divisor": null, "recipe_ranks": [725, 725, 725, 725, 725, 725], "split": "test"}	[11]

Row ID	Run / state	Specification	Split / group / rank scope	Seeds
R011	architecture-replication-20261014-103759-8d32ab84 / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"rank": 725, "rank_divisor": null, "recipe_ranks": [725, 725, 725, 725, 725, 725], "split": "test"}	[11, 23, 37]
R012	architecture-replication-20261014-103759-8d32ab84 / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"rank": 725, "rank_divisor": null, "recipe_ranks": [725, 725, 725, 725, 725, 725], "split": "test"}	[11, 23, 37]
R013	architecture-replication-20261014-103759-8d32ab84 / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"rank": 1024, "rank_divisor": 2, "recipe_ranks": [1024, 1024, 1024, 1024, 1024, 1024], "split": "validation"}	[11, 23, 37]
R014	architecture-replication-20261014-103759-8d32ab84 / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"rank": 1024, "rank_divisor": 2, "recipe_ranks": [1024, 1024, 1024, 1024, 1024, 1024], "split": "validation"}	[11, 23, 37]
R015	architecture-replication-20261014-103759-8d32ab84 / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"rank": 682, "rank_divisor": 3, "recipe_ranks": [682, 682, 682, 682, 682, 682], "split": "validation"}	[11, 23, 37]
R016	architecture-replication-20261014-103759-8d32ab84 / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"rank": 682, "rank_divisor": 3, "recipe_ranks": [682, 682, 682, 682, 682, 682], "split": "validation"}	[11, 23, 37]
R017	architecture-replication-20261014-103759-8d32ab84 / completed	{"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}	{"rank": 512, "rank_divisor": 4, "recipe_ranks": [512, 512, 512, 512, 512, 512], "split": "validation"}	[11, 23, 37]
R018	architecture-replication-20261014-103759-8d32ab84 / completed	{"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}	{"rank": 512, "rank_divisor": 4, "recipe_ranks": [512, 512, 512, 512, 512, 512], "split": "validation"}	[11, 23, 37]

Panel 2 of 3; match by Row ID.

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R001	architecture-replication-20261014-093845-97378b5e / incomplete	2.91722043	unmeasured	0
R002	architecture-replication-20261014-093845-97378b5e / incomplete	3.18099289	unmeasured	0.0616629847
R003	architecture-replication-20261014-093845-97378b5e / incomplete	2.93443868	0.00145544439	0.00615478758
R004	architecture-replication-20261014-093845-97378b5e / incomplete	2.94332386	0.00227178531	0.0127619533
R005	architecture-replication-20261014-093845-97378b5e / incomplete	2.98576098	0.00186708519	0.00527501643
R006	architecture-replication-20261014-093845-97378b5e / incomplete	2.99344161	0.00188966622	0.0110389616
R007	architecture-replication-20261014-093845-97378b5e / incomplete	3.00960245	0.00513867909	0.0111782273
R008	architecture-replication-20261014-093845-97378b5e / incomplete	3.03686124	0.00451877599	0.0223829381
R009	architecture-replication-20261014-103759-8d32ab84 / completed	3.99873556	unmeasured	0
R010	architecture-replication-20261014-103759-8d32ab84 / completed	3.85243198	unmeasured	0.237778731

Row ID	Run / state	NLL mean	NLL seed SD	Drift mean
R011	architecture-replication-20261014-103759-8d32ab84 / completed	4.05640463	0.0228864502	0.0358803942
R012	architecture-replication-20261014-103759-8d32ab84 / completed	3.83591612	0.0103346354	0.13834591
R013	architecture-replication-20261014-103759-8d32ab84 / completed	4.11730048	0.00999990552	0.0131921335
R014	architecture-replication-20261014-103759-8d32ab84 / completed	3.99709851	0.0034649186	0.0500460576
R015	architecture-replication-20261014-103759-8d32ab84 / completed	4.15837853	0.0193157214	0.0388193871
R016	architecture-replication-20261014-103759-8d32ab84 / completed	3.88534704	0.00924738816	0.1558375
R017	architecture-replication-20261014-103759-8d32ab84 / completed	4.22238934	0.0312371256	0.0634802611
R018	architecture-replication-20261014-103759-8d32ab84 / completed	3.89287487	0.00650573852	0.247307618

Panel 3 of 3; match by Row ID.

Row ID	Run / state	Drift seed SD
R001	architecture-replication-20261014-093845-97378b5e / incomplete	unmeasured
R002	architecture-replication-20261014-093845-97378b5e / incomplete	unmeasured
R003	architecture-replication-20261014-093845-97378b5e / incomplete	9.06818768e-05
R004	architecture-replication-20261014-093845-97378b5e / incomplete	0.000160025705
R005	architecture-replication-20261014-093845-97378b5e / incomplete	0.000155706933
R006	architecture-replication-20261014-093845-97378b5e / incomplete	7.87701617e-05
R007	architecture-replication-20261014-093845-97378b5e / incomplete	0.000317143432
R008	architecture-replication-20261014-093845-97378b5e / incomplete	0.000308259687
R009	architecture-replication-20261014-103759-8d32ab84 / completed	unmeasured
R010	architecture-replication-20261014-103759-8d32ab84 / completed	unmeasured
R011	architecture-replication-20261014-103759-8d32ab84 / completed	0.000777615937
R012	architecture-replication-20261014-103759-8d32ab84 / completed	0.000864376837
R013	architecture-replication-20261014-103759-8d32ab84 / completed	0.000154870713
R014	architecture-replication-20261014-103759-8d32ab84 / completed	0.000338835218
R015	architecture-replication-20261014-103759-8d32ab84 / completed	0.000366967559
R016	architecture-replication-20261014-103759-8d32ab84 / completed	0.00119386663

Row ID	Run / state	Drift seed SD
R017	architecture-replication-20261014-103759-8d32ab84 / completed	0.000757422494
R018	architecture-replication-20261014-103759-8d32ab84 / completed	0.00126797867

Recovery provenance: {"independent_replication": false, "interrupted_cost": {"basis": "Conservative actual Python process launch-to-next-system-boot wall interval; includes reboot gap; original status retained unchanged.", "charged_seconds": 3647.1586377620697, "independent_replication": false, "model": "Qwen/Qwen3.5-0.8B", "new_boot_epoch": 1791970132.8027894, "records_saved": 10, "run": "C:\\Users\\NPC\\Desktop\\ResearchCompression\\results\\paper-v2\\architecture-replication-20261014-082807-5c73f669", "run_id": "architecture-replication-20261014-082807-5c73f669", "source_artifacts": {"config.json": "d9a0760000b6e7222b9671a5e0561d61bb9f361355fe777766d5189dcead9898", "contract.json": "703af918c7287806089ad2ec358b9df623af24552ce7b580e0c6fd462a680ab9", "metrics.json": "29422a72a2536d51795bd97abfb112e2a53d2957bed9e0b8be0bcab49aa4c621", "readiness.json": "067bae33ea5bd31a127520d46deadc34a79504cd987a948fd052e7aca31c50f3", "status.json": "2f66cc74973ccbf05b72874c6500e8b742ee6ad8d80770d52b2b81f10d8c32fd", "validation-teacher.json": "057c49abf6c7728756cf3acd2af6f03163b416d9195be9df34fff95cb88aeeb8"}, "stage": "replication", "started_epoch": 1791966485.6441517, "state": "interrupted by reboot"}, "reused_records": 10, "source_artifacts": {"config.json": "d9a0760000b6e7222b9671a5e0561d61bb9f361355fe777766d5189dcead9898", "contract.json": "703af918c7287806089ad2ec358b9df623af24552ce7b580e0c6fd462a680ab9", "metrics.json": "29422a72a2536d51795bd97abfb112e2a53d2957bed9e0b8be0bcab49aa4c621", "readiness.json": "067bae33ea5bd31a127520d46deadc34a79504cd987a948fd052e7aca31c50f3", "status.json": "2f66cc74973ccbf05b72874c6500e8b742ee6ad8d80770d52b2b81f10d8c32fd", "validation-teacher.json": "057c49abf6c7728756cf3acd2af6f03163b416d9195be9df34fff95cb88aeeb8"}, "source_run": "C:\\Users\\NPC\\Desktop\\ResearchCompression\\results\\paper-v2\\architecture-replication-20261014-082807-5c73f669"}. Successor includes preserved source observations; attempts remain separate and are not independent replications.

Scoped finding (architecture-replication-20261014-093845-97378b5e, Qwen/Qwen3.5-0.8B): at the primary test rank, anchoring has higher NLL and higher primary drift than matched whitening in all three calibration seeds. This weakens the anchor hypothesis at this measured scope; it does not establish a universal failure or count reused observations as new replications.

Scoped tradeoff (architecture-replication-20261014-103759-8d32ab84, LiquidAI/LFM2.5-1.2B-Instruct): at the primary test rank, anchoring lowers NLL but increases primary drift versus matched whitening in every historical calibration seed. The likelihood improvement does not establish preserved hidden-state computation or a denoising mechanism. This mixed outcome remains inconclusive under the joint criterion.

Practical costs and calibration

Whole-model checkpoint bytes include archive overhead; factor payload size and whole-model savings are separate quantities. Placements and precision must agree before comparing timings. Below are measured records only; missing conditions remain unmeasured.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 5; match by Row ID.

Row ID	Run / state	Label	Seed	Bits
R001	practical-20261014-113642-0fe078b5 / cancelled	teacher	None	dense / float16
R002	practical-20261014-113642-0fe078b5 / cancelled	teacher	None	dense / float16
R003	practical-20261014-113642-0fe078b5 / cancelled	teacher	None	dense / float16
R004	practical-20261014-113642-0fe078b5 / cancelled	teacher	None	dense / float16
R005	practical-20261014-113642-0fe078b5 / cancelled	baseline	11	16 / float16
R006	practical-20261014-113642-0fe078b5 / cancelled	baseline	11	16 / float16
R007	practical-20261014-113642-0fe078b5 / cancelled	baseline	11	16 / float16

Row ID	Run / state	Label	Seed	Bits
R008	practical-20261014-113642-0fe078b5 / cancelled	baseline	11	16 / float16
R009	practical-20261014-113642-0fe078b5 / cancelled	baseline	23	16 / float16
R010	practical-20261014-113642-0fe078b5 / cancelled	baseline	23	16 / float16
R011	practical-20261014-113642-0fe078b5 / cancelled	baseline	23	16 / float16
R012	practical-20261014-113642-0fe078b5 / cancelled	baseline	23	16 / float16
R013	practical-20261014-113642-0fe078b5 / cancelled	baseline	11	8 / float16
R014	practical-20261014-113642-0fe078b5 / cancelled	baseline	11	8 / float16
R015	practical-20261014-113642-0fe078b5 / cancelled	baseline	11	8 / float16
R016	practical-20261014-113642-0fe078b5 / cancelled	baseline	11	8 / float16
R017	practical-20261014-113642-0fe078b5 / cancelled	baseline	23	8 / float16
R018	practical-20261014-113642-0fe078b5 / cancelled	baseline	23	8 / float16
R019	practical-20261014-113642-0fe078b5 / cancelled	baseline	23	8 / float16
R020	practical-20261014-113642-0fe078b5 / cancelled	baseline	23	8 / float16
R021	practical-20261014-113642-0fe078b5 / cancelled	solver	11	16 / float16
R022	practical-20261014-113642-0fe078b5 / cancelled	solver	11	16 / float16
R023	practical-20261014-113642-0fe078b5 / cancelled	solver	11	16 / float16
R024	practical-20261014-113642-0fe078b5 / cancelled	solver	11	16 / float16
R025	practical-20261014-113642-0fe078b5 / cancelled	solver	11	8 / float16
R026	practical-20261014-113642-0fe078b5 / cancelled	solver	11	8 / float16
R027	practical-20261014-113642-0fe078b5 / cancelled	solver	11	8 / float16
R028	practical-20261014-113642-0fe078b5 / cancelled	solver	11	8 / float16
R029	practical-bf16-20261014-135107-82168ad5 / completed	teacher	None	16 / bfloat16
R030	practical-bf16-20261014-135107-82168ad5 / completed	teacher	None	16 / bfloat16
R031	practical-bf16-20261014-135107-82168ad5 / completed	teacher	None	16 / bfloat16
R032	practical-bf16-20261014-135107-82168ad5 / completed	teacher	None	16 / bfloat16
R033	practical-bf16-20261014-135107-82168ad5 / completed	baseline	11	16 / bfloat16
R034	practical-bf16-20261014-135107-82168ad5 / completed	baseline	11	16 / bfloat16
R035	practical-bf16-20261014-135107-82168ad5 / completed	baseline	11	16 / bfloat16
R036	practical-bf16-20261014-135107-82168ad5 / completed	baseline	11	16 / bfloat16

Row ID	Run / state	Label	Seed	Bits
R037	practical-bf16-20261014-135107-82168ad5 / completed	baseline	23	16 / bfloat16
R038	practical-bf16-20261014-135107-82168ad5 / completed	baseline	23	16 / bfloat16
R039	practical-bf16-20261014-135107-82168ad5 / completed	baseline	23	16 / bfloat16
R040	practical-bf16-20261014-135107-82168ad5 / completed	baseline	23	16 / bfloat16
R041	practical-bf16-20261014-135107-82168ad5 / completed	solver	11	16 / bfloat16
R042	practical-bf16-20261014-135107-82168ad5 / completed	solver	11	16 / bfloat16
R043	practical-bf16-20261014-135107-82168ad5 / completed	solver	11	16 / bfloat16
R044	practical-bf16-20261014-135107-82168ad5 / completed	solver	11	16 / bfloat16
R045	practical-bf16-20261014-135107-82168ad5 / completed	solver	23	16 / bfloat16
R046	practical-bf16-20261014-135107-82168ad5 / completed	solver	23	16 / bfloat16
R047	practical-bf16-20261014-135107-82168ad5 / completed	solver	23	16 / bfloat16
R048	practical-bf16-20261014-135107-82168ad5 / completed	solver	23	16 / bfloat16

Panel 2 of 5; match by Row ID.

Row ID	Run / state	Whole bytes	Factor bytes	Raw NLL
R001	practical-20261014-113642-0fe078b5 / cancelled	1.99988434e+09	unmeasured	3.45156254
R002	practical-20261014-113642-0fe078b5 / cancelled	1.99988434e+09	unmeasured	3.45156254
R003	practical-20261014-113642-0fe078b5 / cancelled	1.99988434e+09	unmeasured	3.45156254
R004	practical-20261014-113642-0fe078b5 / cancelled	1.99988434e+09	unmeasured	3.45156254
R005	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.56275409
R006	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.56275409
R007	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.56275409
R008	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.56275409
R009	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.56985661
R010	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.56985661
R011	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.56985661
R012	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.56985661
R013	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56308141
R014	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56308141
R015	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56308141

Row ID	Run / state	Whole bytes	Factor bytes	Raw NLL
R016	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56308141
R017	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56346292
R018	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56346292
R019	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56346292
R020	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56346292
R021	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.55963483
R022	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.55963483
R023	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.55963483
R024	practical-20261014-113642-0fe078b5 / cancelled	1.98200268e+09	43509058	3.55963483
R025	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56194517
R026	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56194517
R027	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56194517
R028	practical-20261014-113642-0fe078b5 / cancelled	1.96061379e+09	22118386	3.56194517
R029	practical-bf16-20261014-135107-82168ad5 / completed	1.99988371e+09	unmeasured	3.4515495
R030	practical-bf16-20261014-135107-82168ad5 / completed	1.99988371e+09	unmeasured	3.4515495
R031	practical-bf16-20261014-135107-82168ad5 / completed	1.99988371e+09	unmeasured	3.4515495
R032	practical-bf16-20261014-135107-82168ad5 / completed	1.99988371e+09	unmeasured	3.4515495
R033	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.55608618
R034	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.55608618
R035	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.55608618
R036	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.55608618
R037	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.56843147
R038	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.56843147
R039	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.56843147
R040	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.56843147
R041	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.5568263
R042	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.5568263
R043	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.5568263
R044	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.5568263

Row ID	Run / state	Whole bytes	Factor bytes	Raw NLL
R045	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.56294882
R046	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.56294882
R047	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.56294882
R048	practical-bf16-20261014-135107-82168ad5 / completed	1.98200351e+09	43467335	3.56294882

Panel 3 of 5; match by Row ID.

Row ID	Run / state	Scaled NLL	Temperature	Mode
R001	practical-20261014-113642-0fe078b5 / cancelled	3.42255866	1.09050773	prefill
R002	practical-20261014-113642-0fe078b5 / cancelled	3.42255866	1.09050773	decode
R003	practical-20261014-113642-0fe078b5 / cancelled	3.42255866	1.09050773	prefill
R004	practical-20261014-113642-0fe078b5 / cancelled	3.42255866	1.09050773	decode
R005	practical-20261014-113642-0fe078b5 / cancelled	3.54815343	1.09050773	prefill
R006	practical-20261014-113642-0fe078b5 / cancelled	3.54815343	1.09050773	decode
R007	practical-20261014-113642-0fe078b5 / cancelled	3.54815343	1.09050773	prefill
R008	practical-20261014-113642-0fe078b5 / cancelled	3.54815343	1.09050773	decode
R009	practical-20261014-113642-0fe078b5 / cancelled	3.55435184	1.09050773	prefill
R010	practical-20261014-113642-0fe078b5 / cancelled	3.55435184	1.09050773	decode
R011	practical-20261014-113642-0fe078b5 / cancelled	3.55435184	1.09050773	prefill
R012	practical-20261014-113642-0fe078b5 / cancelled	3.55435184	1.09050773	decode
R013	practical-20261014-113642-0fe078b5 / cancelled	3.54883071	1.09050773	prefill
R014	practical-20261014-113642-0fe078b5 / cancelled	3.54883071	1.09050773	decode
R015	practical-20261014-113642-0fe078b5 / cancelled	3.54883071	1.09050773	prefill
R016	practical-20261014-113642-0fe078b5 / cancelled	3.54883071	1.09050773	decode
R017	practical-20261014-113642-0fe078b5 / cancelled	3.54777084	1.09050773	prefill
R018	practical-20261014-113642-0fe078b5 / cancelled	3.54777084	1.09050773	decode
R019	practical-20261014-113642-0fe078b5 / cancelled	3.54777084	1.09050773	prefill
R020	practical-20261014-113642-0fe078b5 / cancelled	3.54777084	1.09050773	decode
R021	practical-20261014-113642-0fe078b5 / cancelled	3.54464574	1.09050773	prefill
R022	practical-20261014-113642-0fe078b5 / cancelled	3.54464574	1.09050773	decode
R023	practical-20261014-113642-0fe078b5 / cancelled	3.54464574	1.09050773	prefill

Row ID	Run / state	Scaled NLL	Temperature	Mode
R024	practical-20261014-113642-0fe078b5 / cancelled	3.54464574	1.09050773	decode
R025	practical-20261014-113642-0fe078b5 / cancelled	3.54677328	1.09050773	prefill
R026	practical-20261014-113642-0fe078b5 / cancelled	3.54677328	1.09050773	decode
R027	practical-20261014-113642-0fe078b5 / cancelled	3.54677328	1.09050773	prefill
R028	practical-20261014-113642-0fe078b5 / cancelled	3.54677328	1.09050773	decode
R029	practical-bf16-20261014-135107-82168ad5 / completed	3.42260056	1.09050773	prefill
R030	practical-bf16-20261014-135107-82168ad5 / completed	3.42260056	1.09050773	decode
R031	practical-bf16-20261014-135107-82168ad5 / completed	3.42260056	1.09050773	prefill
R032	practical-bf16-20261014-135107-82168ad5 / completed	3.42260056	1.09050773	decode
R033	practical-bf16-20261014-135107-82168ad5 / completed	3.54172555	1.09050773	prefill
R034	practical-bf16-20261014-135107-82168ad5 / completed	3.54172555	1.09050773	decode
R035	practical-bf16-20261014-135107-82168ad5 / completed	3.54172555	1.09050773	prefill
R036	practical-bf16-20261014-135107-82168ad5 / completed	3.54172555	1.09050773	decode
R037	practical-bf16-20261014-135107-82168ad5 / completed	3.55326688	1.09050773	prefill
R038	practical-bf16-20261014-135107-82168ad5 / completed	3.55326688	1.09050773	decode
R039	practical-bf16-20261014-135107-82168ad5 / completed	3.55326688	1.09050773	prefill
R040	practical-bf16-20261014-135107-82168ad5 / completed	3.55326688	1.09050773	decode
R041	practical-bf16-20261014-135107-82168ad5 / completed	3.5422546	1.09050773	prefill
R042	practical-bf16-20261014-135107-82168ad5 / completed	3.5422546	1.09050773	decode
R043	practical-bf16-20261014-135107-82168ad5 / completed	3.5422546	1.09050773	prefill
R044	practical-bf16-20261014-135107-82168ad5 / completed	3.5422546	1.09050773	decode
R045	practical-bf16-20261014-135107-82168ad5 / completed	3.54534144	1.09050773	prefill
R046	practical-bf16-20261014-135107-82168ad5 / completed	3.54534144	1.09050773	decode
R047	practical-bf16-20261014-135107-82168ad5 / completed	3.54534144	1.09050773	prefill
R048	practical-bf16-20261014-135107-82168ad5 / completed	3.54534144	1.09050773	decode

Panel 4 of 5; match by Row ID.

Row ID	Run / state	Prefill	Median s	p95 s
R001	practical-20261014-113642-0fe078b5 / cancelled	128	0.05783475	0.059800255
R002	practical-20261014-113642-0fe078b5 / cancelled	128	6.4778003	6.64036795

Row ID	Run / state	Prefill	Median s	p95 s
R003	practical-20261014-113642-0fe078b5 / cancelled	512	0.055825	0.05969351
R004	practical-20261014-113642-0fe078b5 / cancelled	512	6.05921945	6.1403141
R005	practical-20261014-113642-0fe078b5 / cancelled	128	0.0590297	0.08538865
R006	practical-20261014-113642-0fe078b5 / cancelled	128	6.44498855	6.62319019
R007	practical-20261014-113642-0fe078b5 / cancelled	512	0.05369865	0.055011425
R008	practical-20261014-113642-0fe078b5 / cancelled	512	6.11049025	6.4726321
R009	practical-20261014-113642-0fe078b5 / cancelled	128	0.05889345	0.06917032
R010	practical-20261014-113642-0fe078b5 / cancelled	128	6.59023655	6.83514165
R011	practical-20261014-113642-0fe078b5 / cancelled	512	0.05657515	0.06035711
R012	practical-20261014-113642-0fe078b5 / cancelled	512	6.14512885	6.47288771
R013	practical-20261014-113642-0fe078b5 / cancelled	128	0.06479795	0.09071199
R014	practical-20261014-113642-0fe078b5 / cancelled	128	7.11784765	7.23628017
R015	practical-20261014-113642-0fe078b5 / cancelled	512	0.0592873	0.06291703
R016	practical-20261014-113642-0fe078b5 / cancelled	512	6.7235441	6.79397279
R017	practical-20261014-113642-0fe078b5 / cancelled	128	0.0648798	0.066804915
R018	practical-20261014-113642-0fe078b5 / cancelled	128	7.2303226	8.77872235
R019	practical-20261014-113642-0fe078b5 / cancelled	512	0.0611183	0.065548205
R020	practical-20261014-113642-0fe078b5 / cancelled	512	7.09512565	7.33472052
R021	practical-20261014-113642-0fe078b5 / cancelled	128	0.0603992	0.072276215
R022	practical-20261014-113642-0fe078b5 / cancelled	128	6.6212441	6.8661912
R023	practical-20261014-113642-0fe078b5 / cancelled	512	0.0581729	0.06183209
R024	practical-20261014-113642-0fe078b5 / cancelled	512	6.18178745	6.36525901
R025	practical-20261014-113642-0fe078b5 / cancelled	128	0.06678435	0.072467045
R026	practical-20261014-113642-0fe078b5 / cancelled	128	7.19626915	7.34018282
R027	practical-20261014-113642-0fe078b5 / cancelled	512	0.06112195	0.064887285
R028	practical-20261014-113642-0fe078b5 / cancelled	512	6.7455395	6.95685991
R029	practical-bf16-20261014-135107-82168ad5 / completed	128	0.05917745	0.070606515
R030	practical-bf16-20261014-135107-82168ad5 / completed	128	6.42761055	6.498819
R031	practical-bf16-20261014-135107-82168ad5 / completed	512	0.0544619	0.057006585

Row ID	Run / state	Prefill	Median s	p95 s
R032	practical-bf16-20261014-135107-82168ad5 / completed	512	6.23879075	6.43212847
R033	practical-bf16-20261014-135107-82168ad5 / completed	128	0.0620393	0.072688375
R034	practical-bf16-20261014-135107-82168ad5 / completed	128	6.65239975	6.93452725
R035	practical-bf16-20261014-135107-82168ad5 / completed	512	0.0567413	0.06545248
R036	practical-bf16-20261014-135107-82168ad5 / completed	512	6.3900002	6.60615606
R037	practical-bf16-20261014-135107-82168ad5 / completed	128	0.06372255	0.07085632
R038	practical-bf16-20261014-135107-82168ad5 / completed	128	6.6857301	7.01892626
R039	practical-bf16-20261014-135107-82168ad5 / completed	512	0.05767275	0.05970958
R040	practical-bf16-20261014-135107-82168ad5 / completed	512	6.40245935	6.74076871
R041	practical-bf16-20261014-135107-82168ad5 / completed	128	0.07367855	0.08038677
R042	practical-bf16-20261014-135107-82168ad5 / completed	128	7.36599745	8.0900931
R043	practical-bf16-20261014-135107-82168ad5 / completed	512	0.0573537	0.0659666
R044	practical-bf16-20261014-135107-82168ad5 / completed	512	6.31099465	6.43883127
R045	practical-bf16-20261014-135107-82168ad5 / completed	128	0.05970755	0.06792561
R046	practical-bf16-20261014-135107-82168ad5 / completed	128	6.4710207	6.57088731
R047	practical-bf16-20261014-135107-82168ad5 / completed	512	0.05463395	0.056230945
R048	practical-bf16-20261014-135107-82168ad5 / completed	512	6.1332589	6.25817783

Panel 5 of 5; match by Row ID.

Row ID	Run / state	Allocated bytes	Reserved bytes
R001	practical-20261014-113642-0fe078b5 / cancelled	2.13196698e+09	2.1495808e+09
R002	practical-20261014-113642-0fe078b5 / cancelled	2.07219046e+09	2.15587226e+09
R003	practical-20261014-113642-0fe078b5 / cancelled	2.34440499e+09	2.40543334e+09
R004	practical-20261014-113642-0fe078b5 / cancelled	2.08504525e+09	2.40543334e+09
R005	practical-20261014-113642-0fe078b5 / cancelled	2.1154519e+09	2.19571814e+09
R006	practical-20261014-113642-0fe078b5 / cancelled	2.05567539e+09	2.1978153e+09
R007	practical-20261014-113642-0fe078b5 / cancelled	2.32788992e+09	2.44527923e+09
R008	practical-20261014-113642-0fe078b5 / cancelled	2.06859571e+09	2.44527923e+09
R009	practical-20261014-113642-0fe078b5 / cancelled	2.1154519e+09	2.19571814e+09
R010	practical-20261014-113642-0fe078b5 / cancelled	2.05567539e+09	2.1978153e+09

Row ID	Run / state	Allocated bytes	Reserved bytes
R011	practical-20261014-113642-0fe078b5 / cancelled	2.32788992e+09	2.44527923e+09
R012	practical-20261014-113642-0fe078b5 / cancelled	2.06859571e+09	2.44527923e+09
R013	practical-20261014-113642-0fe078b5 / cancelled	2.09404621e+09	2.16426086e+09
R014	practical-20261014-113642-0fe078b5 / cancelled	2.03845376e+09	2.16845517e+09
R015	practical-20261014-113642-0fe078b5 / cancelled	2.30648422e+09	2.42011341e+09
R016	practical-20261014-113642-0fe078b5 / cancelled	2.04719002e+09	2.42011341e+09
R017	practical-20261014-113642-0fe078b5 / cancelled	2.09404621e+09	2.16426086e+09
R018	practical-20261014-113642-0fe078b5 / cancelled	2.03845376e+09	2.16845517e+09
R019	practical-20261014-113642-0fe078b5 / cancelled	2.30648422e+09	2.42011341e+09
R020	practical-20261014-113642-0fe078b5 / cancelled	2.04719002e+09	2.42011341e+09
R021	practical-20261014-113642-0fe078b5 / cancelled	2.1154519e+09	2.19571814e+09
R022	practical-20261014-113642-0fe078b5 / cancelled	2.05567539e+09	2.1978153e+09
R023	practical-20261014-113642-0fe078b5 / cancelled	2.32788992e+09	2.44527923e+09
R024	practical-20261014-113642-0fe078b5 / cancelled	2.06859571e+09	2.44527923e+09
R025	practical-20261014-113642-0fe078b5 / cancelled	2.09404621e+09	2.16426086e+09
R026	practical-20261014-113642-0fe078b5 / cancelled	2.03845376e+09	2.16845517e+09
R027	practical-20261014-113642-0fe078b5 / cancelled	2.30648422e+09	2.42011341e+09
R028	practical-20261014-113642-0fe078b5 / cancelled	2.04719002e+09	2.42011341e+09
R029	practical-bf16-20261014-135107-82168ad5 / completed	2.13196698e+09	2.1495808e+09
R030	practical-bf16-20261014-135107-82168ad5 / completed	2.07219046e+09	2.15587226e+09
R031	practical-bf16-20261014-135107-82168ad5 / completed	2.34440499e+09	2.40543334e+09
R032	practical-bf16-20261014-135107-82168ad5 / completed	2.08504525e+09	2.40543334e+09
R033	practical-bf16-20261014-135107-82168ad5 / completed	2.12226765e+09	2.1537751e+09
R034	practical-bf16-20261014-135107-82168ad5 / completed	2.06249114e+09	2.15587226e+09
R035	practical-bf16-20261014-135107-82168ad5 / completed	2.33470566e+09	2.38865613e+09
R036	practical-bf16-20261014-135107-82168ad5 / completed	2.07541146e+09	2.38865613e+09
R037	practical-bf16-20261014-135107-82168ad5 / completed	2.12226765e+09	2.1537751e+09
R038	practical-bf16-20261014-135107-82168ad5 / completed	2.06249114e+09	2.15587226e+09
R039	practical-bf16-20261014-135107-82168ad5 / completed	2.33470566e+09	2.38865613e+09

Row ID	Run / state	Allocated bytes	Reserved bytes
R040	practical-bf16-20261014-135107-82168ad5 / completed	2.07541146e+09	2.38865613e+09
R041	practical-bf16-20261014-135107-82168ad5 / completed	2.12226765e+09	2.1537751e+09
R042	practical-bf16-20261014-135107-82168ad5 / completed	2.06249114e+09	2.15587226e+09
R043	practical-bf16-20261014-135107-82168ad5 / completed	2.33470566e+09	2.38865613e+09
R044	practical-bf16-20261014-135107-82168ad5 / completed	2.07541146e+09	2.38865613e+09
R045	practical-bf16-20261014-135107-82168ad5 / completed	2.12226765e+09	2.1537751e+09
R046	practical-bf16-20261014-135107-82168ad5 / completed	2.06249114e+09	2.15587226e+09
R047	practical-bf16-20261014-135107-82168ad5 / completed	2.33470566e+09	2.38865613e+09
R048	practical-bf16-20261014-135107-82168ad5 / completed	2.07541146e+09	2.38865613e+09

Practical measurement scope

The historical FP16/INT8 stage's hardware-gates.json, when present, uses the same seed's FP16 whitening reference for both FP16 and INT8 solver factors. Its threshold concerns median latency and peak allocated memory only; it is not a same-bit INT8 gate and does not test p95, reserved memory, serialized bytes or quality. The later user-authorized BF16-only cohort is separate: seeds 11/23, five expected conditions, no other model formats and no three-seed promotion claim. BF16 has 16 storage bits but must not be labeled FP16. Its calibration routine saves full-target aggregate NLL and ECE, not per-block losses; practical NLL points are descriptive, without paired block intervals. Historical main-quality cohorts retain their separate block uncertainty. The separate descriptive ratios below use same-run, same-seed, same-bit whitening/anchor records with matching device, declared placement, compute precision and workload. Missing or ambiguous pairs remain excluded in report.json. No original hardware gate is relabeled.

Timings use batch size 1, 10 warmups and 30 synchronized samples. Cached 128-token decode timing excludes the initial prompt pass; prefill is measured separately at 128 and 512 tokens. Peak allocated/reserved bytes are CUDA allocator statistics including resident model and cache, not total driver GPU memory or system RAM. INT8 factors are dequantized for floating-point multiplication, so these timings do not establish native INT8-kernel acceleration.

Whole-model reload parity covers a 128-token prompt and one cached next-token forward, not the complete evaluation corpus or arbitrary generation. The historical dense-teacher file lacks this probe; the BF16-only amendment probes its dense teacher as well as its compressed models. Historical local factor probes and BF16 whole-model probes are distinct. The practical stage cap is cooperative; terminal elapsed wall time remains the charged cost.

Same-bit descriptive cost ratios

Each ratio is anchor divided by matched whitening; below 1 is lower cost. Placement is explicit row metadata where available, otherwise the full-model resident device declared in the same-run configuration. These are exploratory measured workload comparisons, not new promotion gates.

Wide-table continuation: identical row IDs identify the same original record across every panel. IDs restart for each source table.

Panel 1 of 4; match by Row ID.

Row ID	Run	Seed	Factor bits	Device / placement
R001	practical-20261014-113642-0fe078b5	11	16	cuda / "resident"
R002	practical-20261014-113642-0fe078b5	11	16	cuda / "resident"
R003	practical-20261014-113642-0fe078b5	11	16	cuda / "resident"
R004	practical-20261014-113642-0fe078b5	11	16	cuda / "resident"

Row ID	Run	Seed	Factor bits	Device / placement
R005	practical-20261014-113642-0fe078b5	11	8	cuda / "resident"
R006	practical-20261014-113642-0fe078b5	11	8	cuda / "resident"
R007	practical-20261014-113642-0fe078b5	11	8	cuda / "resident"
R008	practical-20261014-113642-0fe078b5	11	8	cuda / "resident"
R009	practical-bf16-20261014-135107-82168ad5	11	16	cuda / "resident"
R010	practical-bf16-20261014-135107-82168ad5	11	16	cuda / "resident"
R011	practical-bf16-20261014-135107-82168ad5	11	16	cuda / "resident"
R012	practical-bf16-20261014-135107-82168ad5	11	16	cuda / "resident"
R013	practical-bf16-20261014-135107-82168ad5	23	16	cuda / "resident"
R014	practical-bf16-20261014-135107-82168ad5	23	16	cuda / "resident"
R015	practical-bf16-20261014-135107-82168ad5	23	16	cuda / "resident"
R016	practical-bf16-20261014-135107-82168ad5	23	16	cuda / "resident"

Panel 2 of 4; match by Row ID.

Row ID	Run	Mode	Prefill	Decode
R001	practical-20261014-113642-0fe078b5	prefill	128	0
R002	practical-20261014-113642-0fe078b5	decode	128	128
R003	practical-20261014-113642-0fe078b5	prefill	512	0
R004	practical-20261014-113642-0fe078b5	decode	512	128
R005	practical-20261014-113642-0fe078b5	prefill	128	0
R006	practical-20261014-113642-0fe078b5	decode	128	128
R007	practical-20261014-113642-0fe078b5	prefill	512	0
R008	practical-20261014-113642-0fe078b5	decode	512	128
R009	practical-bf16-20261014-135107-82168ad5	prefill	128	0
R010	practical-bf16-20261014-135107-82168ad5	decode	128	128
R011	practical-bf16-20261014-135107-82168ad5	prefill	512	0
R012	practical-bf16-20261014-135107-82168ad5	decode	512	128
R013	practical-bf16-20261014-135107-82168ad5	prefill	128	0
R014	practical-bf16-20261014-135107-82168ad5	decode	128	128
R015	practical-bf16-20261014-135107-82168ad5	prefill	512	0

Row ID	Run	Mode	Prefill	Decode
R016	practical-bf16-20261014-135107-82168ad5	decode	512	128

Panel 3 of 4; match by Row ID.

Row ID	Run	Median ratio	p95 ratio	Allocated ratio
R001	practical-20261014-113642-0fe078b5	1.02320019	0.846438198	1
R002	practical-20261014-113642-0fe078b5	1.02734769	1.03668942	1
R003	practical-20261014-113642-0fe078b5	1.08332146	1.12398634	1
R004	practical-20261014-113642-0fe078b5	1.011668	0.983411217	1
R005	practical-20261014-113642-0fe078b5	1.03065529	0.798869532	1
R006	practical-20261014-113642-0fe078b5	1.01101759	1.01435857	1
R007	practical-20261014-113642-0fe078b5	1.03094508	1.03131513	1
R008	practical-20261014-113642-0fe078b5	1.0032714	1.02397524	1
R009	practical-bf16-20261014-135107-82168ad5	1.18761092	1.10590958	1
R010	practical-bf16-20261014-135107-82168ad5	1.10726922	1.16663946	1
R011	practical-bf16-20261014-135107-82168ad5	1.01079284	1.00785486	1
R012	practical-bf16-20261014-135107-82168ad5	0.987636065	0.974671384	1
R013	practical-bf16-20261014-135107-82168ad5	0.936992478	0.958638693	1
R014	practical-bf16-20261014-135107-82168ad5	0.967885422	0.936167024	1
R015	practical-bf16-20261014-135107-82168ad5	0.947309605	0.941740756	1
R016	practical-bf16-20261014-135107-82168ad5	0.957953587	0.928407146	1

Panel 4 of 4; match by Row ID.

Row ID	Run	Reserved ratio	Factor-byte ratio	Whole-byte ratio
R001	practical-20261014-113642-0fe078b5	1	1	1
R002	practical-20261014-113642-0fe078b5	1	1	1
R003	practical-20261014-113642-0fe078b5	1	1	1
R004	practical-20261014-113642-0fe078b5	1	1	1
R005	practical-20261014-113642-0fe078b5	1	1	1
R006	practical-20261014-113642-0fe078b5	1	1	1
R007	practical-20261014-113642-0fe078b5	1	1	1
R008	practical-20261014-113642-0fe078b5	1	1	1

Row ID	Run	Reserved ratio	Factor-byte ratio	Whole-byte ratio
R009	practical-bf16-20261014-135107-82168ad5	1	1	1
R010	practical-bf16-20261014-135107-82168ad5	1	1	1
R011	practical-bf16-20261014-135107-82168ad5	1	1	1
R012	practical-bf16-20261014-135107-82168ad5	1	1	1
R013	practical-bf16-20261014-135107-82168ad5	1	1	1
R014	practical-bf16-20261014-135107-82168ad5	1	1	1
R015	practical-bf16-20261014-135107-82168ad5	1	1	1
R016	practical-bf16-20261014-135107-82168ad5	1	1	1

Statistical interpretation

NLL differences use 2,000 paired bootstrap resamples of fixed contiguous 4,096-target blocks. The final partial block remains in full-test NLL but is excluded from fixed-block resampling. Individual dependent tokens are not bootstrapped. Per-seed confidence intervals, exact excluded counts, matched target checks and all scoped candidate-minus-baseline differences are in report.json and report.md. These are evaluation-block intervals, not seed intervals.

Limitations

WikiText-2 has already informed this project, so these results are regression evidence rather than fresh-domain confirmation. No downstream-task capability or arbitrary-generation equivalence was measured. Only attention-output projections are compressed; hybrid DeltaNet/convolution mixers and all MLPs remain dense.

Within-model comparisons match factor budgets and native evaluation targets. Differences across model cohorts cannot be attributed solely to architecture: pretrained weights, tokenizers and, for the historical Gemma reference, inference precision differ. Raw perplexities must not rank different tokenizers. The later BF16 Gemma practical cohort measures costs and calibration; it does not add new BF16 drift evidence.

The original strict numerical-readiness failure remains binding. The two-seed BF16 practical amendment cannot satisfy a three-seed replication gate. Residual drift is a diagnostic proxy, not a mathematical certificate of global computation; lower perplexity on one corpus likewise does not establish preservation. The joint criterion is an operational research gate, not a universal theorem.

Weight-count reduction, actual serialized bytes and measured runtime are separate outcomes. Timings are descriptive for the recorded GPU/software/workloads, and reload probes cover only the stated prompt and cached token. Budget-limited frontier points remain unmeasured. Reboot accounting, including an unbounded first-reboot log tail, prevents an exact 24 GPU-hour compliance claim.

Future scope: preparation and runtime optimization

Future work should separate the cost of constructing compressed models from the cost of running them. The directions below are proposals, not measured speedups or authorization for additional experiments. BF16 should remain the initial execution format, with at most two fixed calibration seeds under the current user constraint; any broader replication requires a separately agreed protocol and budget.

Faster offline compression

Profile activation capture, covariance construction, decomposition, factor transfer and checkpoint serialization separately. Cache reusable teacher activations and sufficient statistics with keys binding the exact model revision, target, token positions, calibration seed and dtype. Reuse must preserve the fitting objective and split boundaries. Batched preparation or approximate decompositions could then be evaluated against the current solver for preparation time, peak CPU/GPU memory, reconstruction error and held-out global behavior; they should not be assumed numerically equivalent.

Efficient factorized inference

The deployed projection computes z=Vx followed by y=Uz, with the original bias applied once. A future BF16 implementation could investigate fused or tiled execution that avoids writing the complete rank-sized intermediate tensor to device memory, reduces launch overhead and uses suitable factor layouts. Compilation, reusable execution graphs and reusable buffers are additional candidates to profile. These changes must retain the compact factors; reconstructing and storing the full dense matrix would surrender the targeted weight-storage reduction. Fewer weights alone do not guarantee lower latency, since the factorized path introduces a dependent operation and its performance depends on shapes and workloads.

Runtime-aware rank and placement choices

A separately frozen validation study could allocate ranks across projections using both error sensitivity and measured BF16 kernel cost, rather than assuming one uniform rank is optimal. It should compare against a matched total parameter or byte budget, keep test data untouched during selection, and retain a strong whitening baseline. GPU-resident and RAM-offloaded execution should be reported as separate conditions: host-device transfers, placement and intermediate buffers belong in the end-to-end cost. Extending compression to MLPs or hybrid non-attention mixers remains a separate scientific question.

Verification and decision rules

For each implementation change, first check numerical agreement with the existing BF16 factorized operator and then evaluate full-model likelihood, logits and drift. Measure offline preparation and online inference separately, including prefill, cached decode, multiple prompt lengths and batch sizes, synchronized median/p95 latency, tokens per second, serialized bytes, allocator memory and independently measured total device/system memory. Compare the optimized factors with both the current factorized implementation and the dense BF16 teacher under identical workload and placement. Report speedups only where observed, alongside quality changes and uncertainty. More extensive reload/generation probes, a fresh pinned corpus and downstream tasks could test whether any benefit generalizes; preservation of global computation remains unresolved.

Reproducibility and incomplete-run appendix

The accompanying report.json embeds each retained contract, source hashes, package/device versions, asset revisions, calibration/split hashes, target recipes, run-specific command, raw per-seed metrics, integrity checks, artifact paths and all failed/interrupted attempts. Read errors are retained explicitly; promotion is disabled.

Stage	Recorded charged wall seconds	Declared cap seconds	Unknown-cost attempts
readiness	1381.392	3600	[]
regularization	1589.148	14400	[]
baselines	3736.553	25200	[]
composition	12825.574	14400	[]
architecture	10720.138	14400	[]
practical	11524.515	14400	[]

Preserved interrupted attempts and reused observations

The main architecture table and figures show the unique validated continuation, including its actual incomplete/terminal state. The source observations are identical reused measurements, not new replications. Original attempts and full metrics remain separately recorded in report.json and report.md; all source hashes and validation warnings are in presentation_supersession.

Original attempt / state	Continuation / state	Reused rows	Original coverage	Charged wall seconds
architecture-replication-20261014-082807-5c73f669 / interrupted	architecture-replication-20261014-093845-97378b5e / incomplete	10	{"frontier_validation_records": 2, "note": "Record-count checks do not replace seed, target, recipe and provenance checks; an earlier 8-record cohort has no amended frontier evidence.", "primary_test_records": 8, "record_counts_complete": false, "required_frontier_validation_records": 18, "required_primary_test_records": 8, "validation_rank_divisors": [2]}	3647.15864

Source provenance (architecture-replication-20261014-082807-5c73f669): {"config.json": "d9a0760000b6e7222b9671a5e0561d61bb9f361355fe777766d5189dcead9898", "contract.json": "703af918c7287806089ad2ec358b9df623af24552ce7b580e0c6fd462a680ab9", "metrics.json": "29422a72a2536d51795bd97abfb112e2a53d2957bed9e0b8be0bcab49aa4c621", "readiness.json": "067bae33ea5bd31a127520d46deadc34a79504cd987a948fd052e7aca31c50f3", "status.json": "2f66cc74973ccbf05b72874c6500e8b742ee6ad8d80770d52b2b81f10d8c32fd", "validation-teacher.json": "057c49abf6c7728756cf3acd2af6f03163b416d9195be9df34fff95cb88aeeb8"}.

Elapsed-wall-time charges include failed/retried work and may include reboot downtime. The first reboot has an unbounded unlogged tail; no exact GPU-hour compliance claim.

Conclusion

Low-rank factors reduce targeted weight counts. Anchoring has no established consistent advantage over matched whitening on both likelihood and representation drift; reliable preservation of global computation remains unresolved. Recorded storage and runtime tradeoffs are reported separately, and all findings remain exploratory.

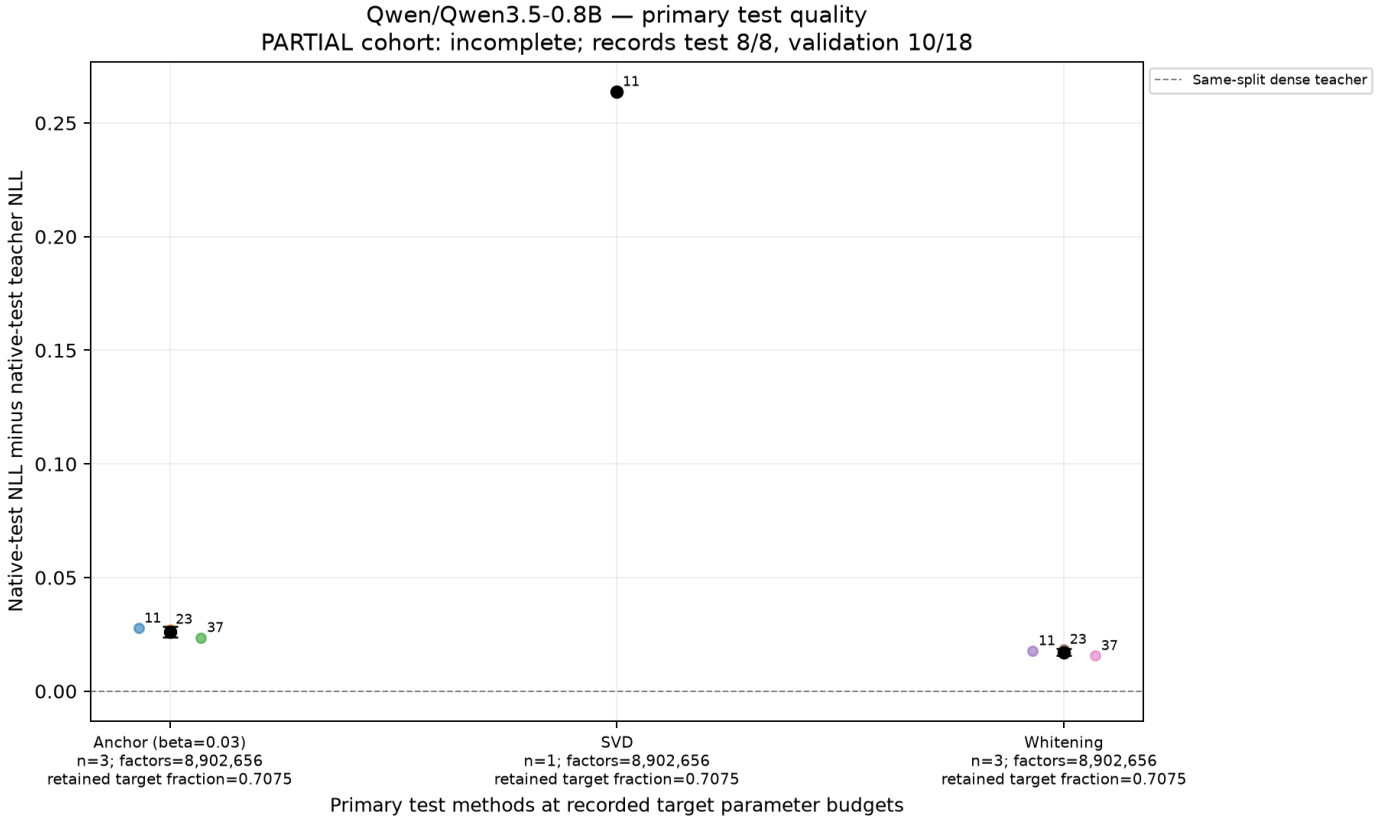
Complete negative or mixed comparisons are retained as bounded evidence. Unmeasured downstream tasks, fresh-corpus confirmation, absent seeds or hardware conditions are not implied by dispatcher completion.

Verified bibliography and dates

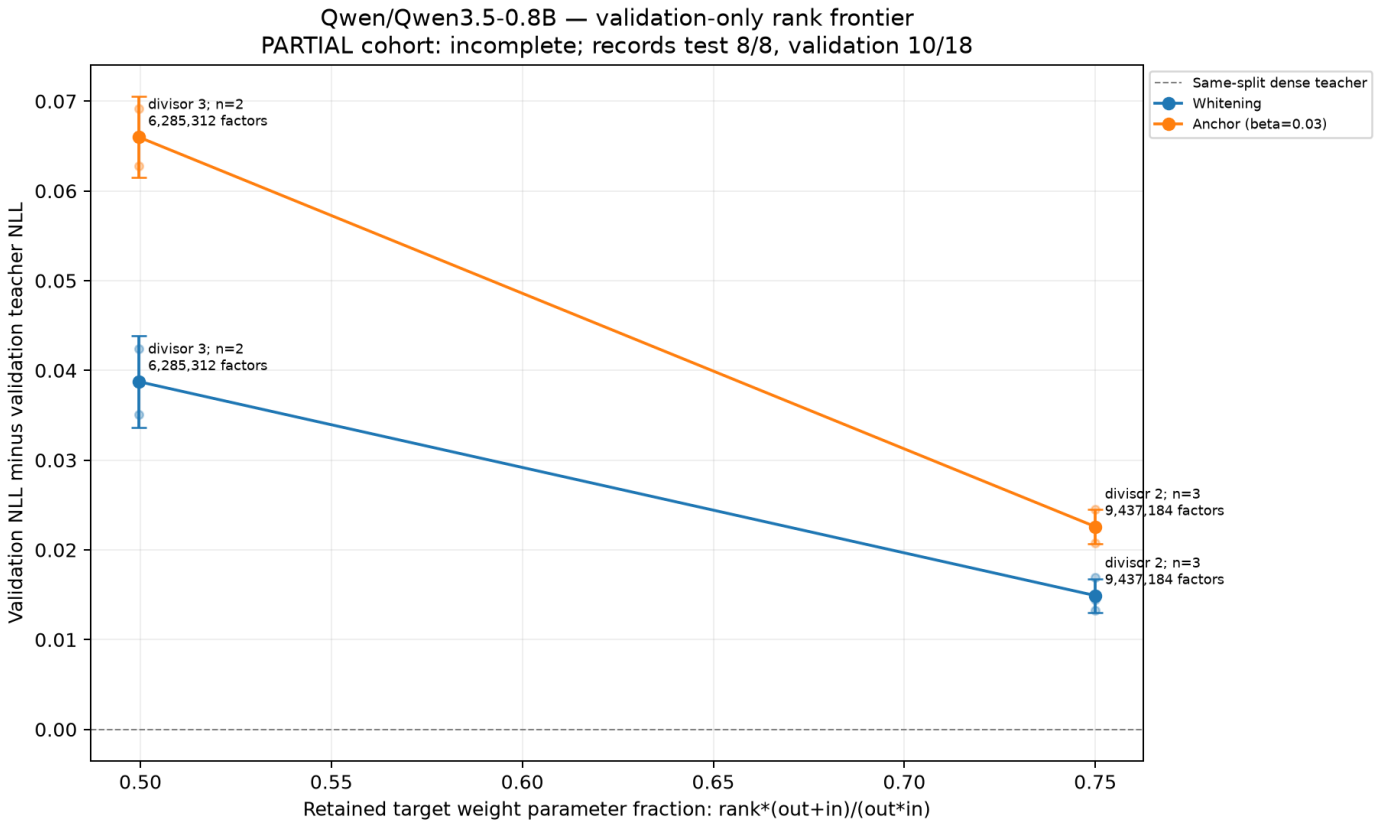
The additive BibTeX file (paper_protocol_v2.bib) contains the 22 plan references, verified from primary metadata on 2026-10-13. Preprint entries use the first submission year, record the checked revision, and separately note a venue when the primary page identifies one. In particular SVD-LLM is a 2024 preprint with ICLR 2025 metadata, GPTQ is a 2022 preprint with ICLR 2023 metadata, AWQ is a 2023 preprint with MLSys 2024 metadata, and SVDQuant is a 2024 preprint with ICLR 2025 metadata. EoRA's current primary metadata identifies ICLR 2026 workshops, not main-conference publication. T5 is cited as its 2020 JMLR article, with its 2019 preprint date recorded separately. Unverified venues are omitted.

Author lists reflect the verified source version. The ASVD v5 author list and the EoRA v6 list differ from some earlier bibliographies; this file does not silently mix versions. Hsu et al.'s author spelling follows the conference PDF (Sung-En Chang); the arXiv metadata spells that name Sungen Chang. The direct OpenReview page returned a browser challenge, so metadata was cross-checked with the primary paper's indexed PDF and its arXiv record.

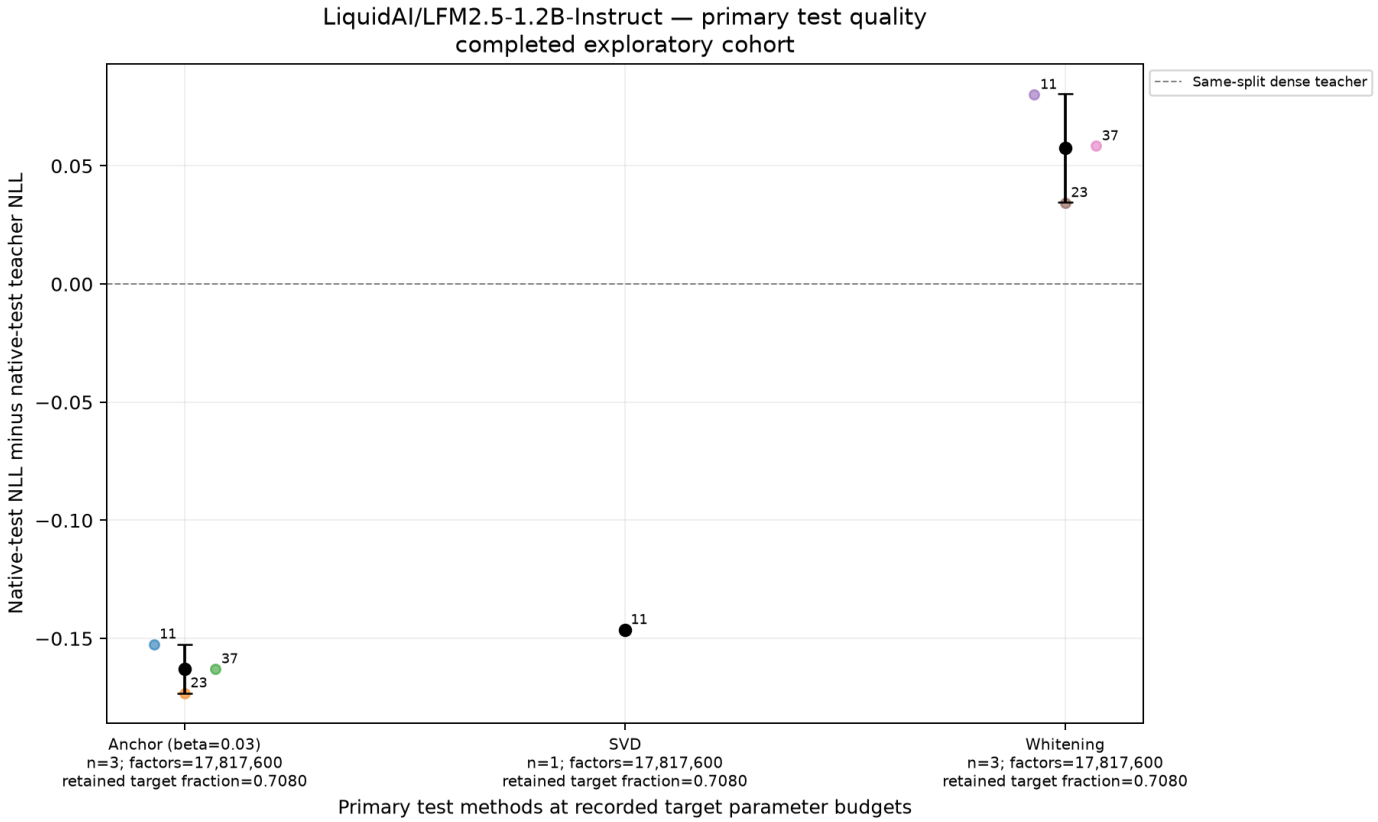
Measured scientific figures



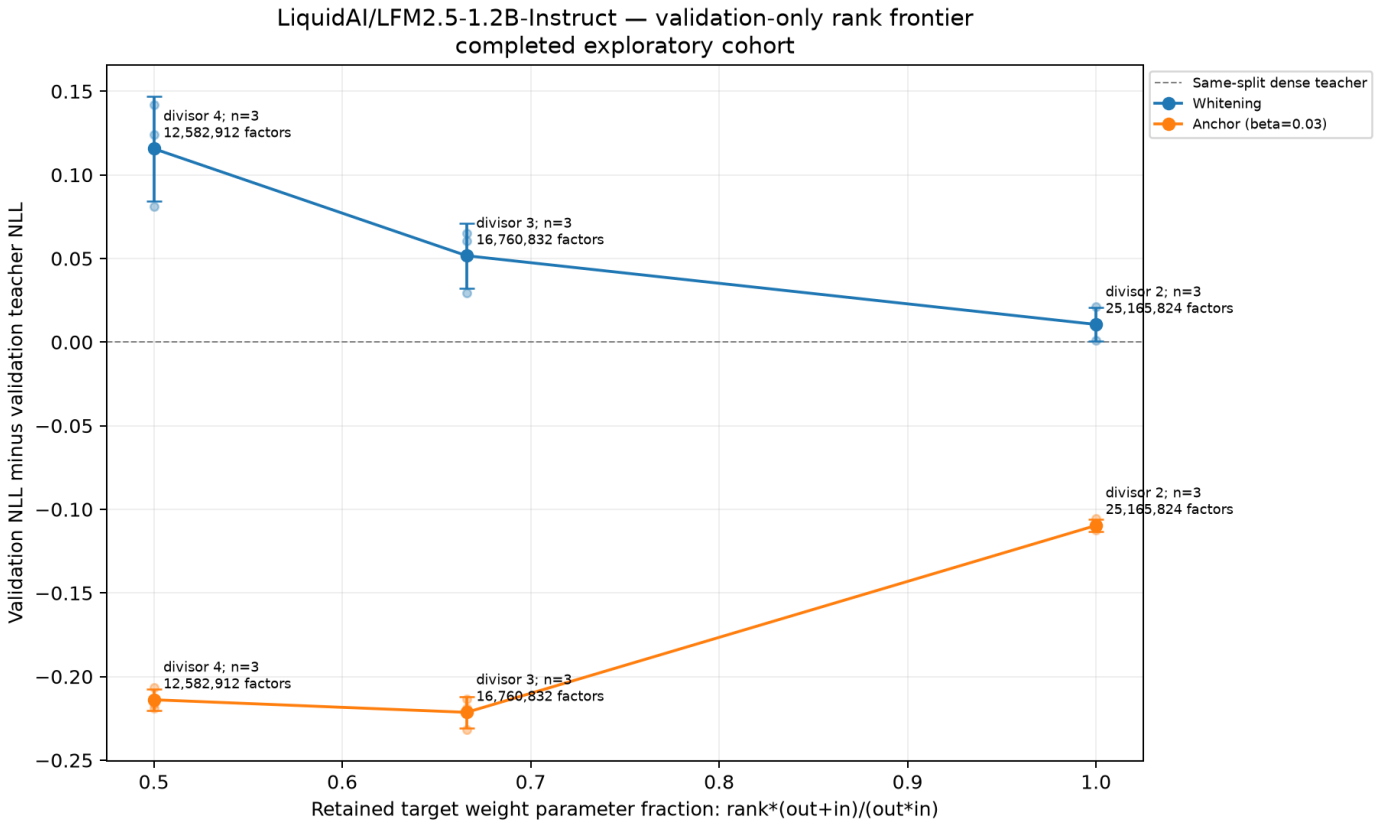
Qwen/Qwen3.5-0.8B: primary native-test teacher-relative NLL for PARTIAL cohort: incomplete; records test 8/8, validation 10/18; individual paired seed points, mean and sample seed SD when at least two seeds exist. Actual target factor counts/fractions shown. Strict promotion remains disabled.



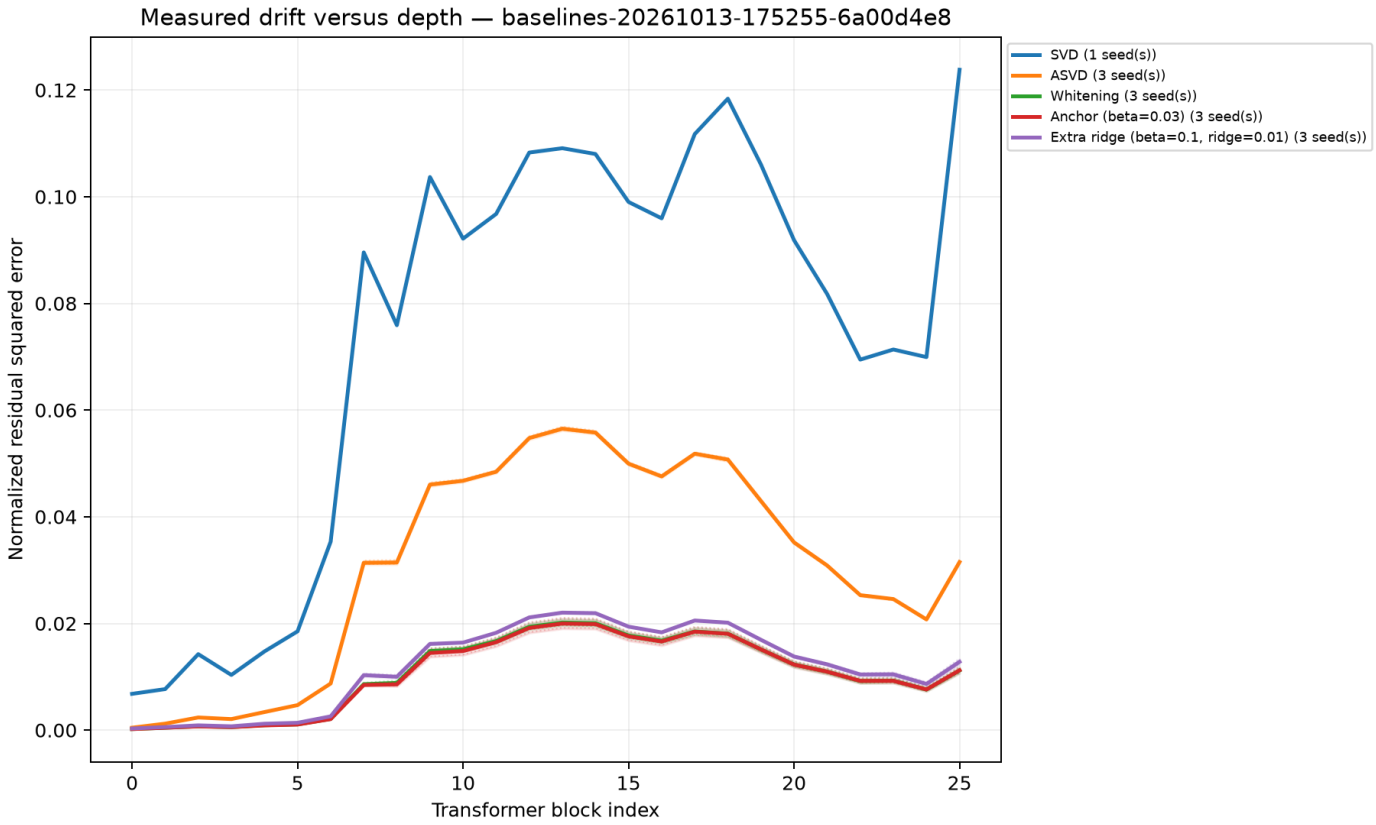
Qwen/Qwen3.5-0.8B: validation-only rank frontier for PARTIAL cohort: incomplete; records test 8/8, validation 10/18. Each point aggregates only one exact rank recipe and available paired seeds; error bars are sample seed SD. X is actual retained target weight parameter fraction, not whole-model storage or a rank compression factor. Validation is not independent confirmation.



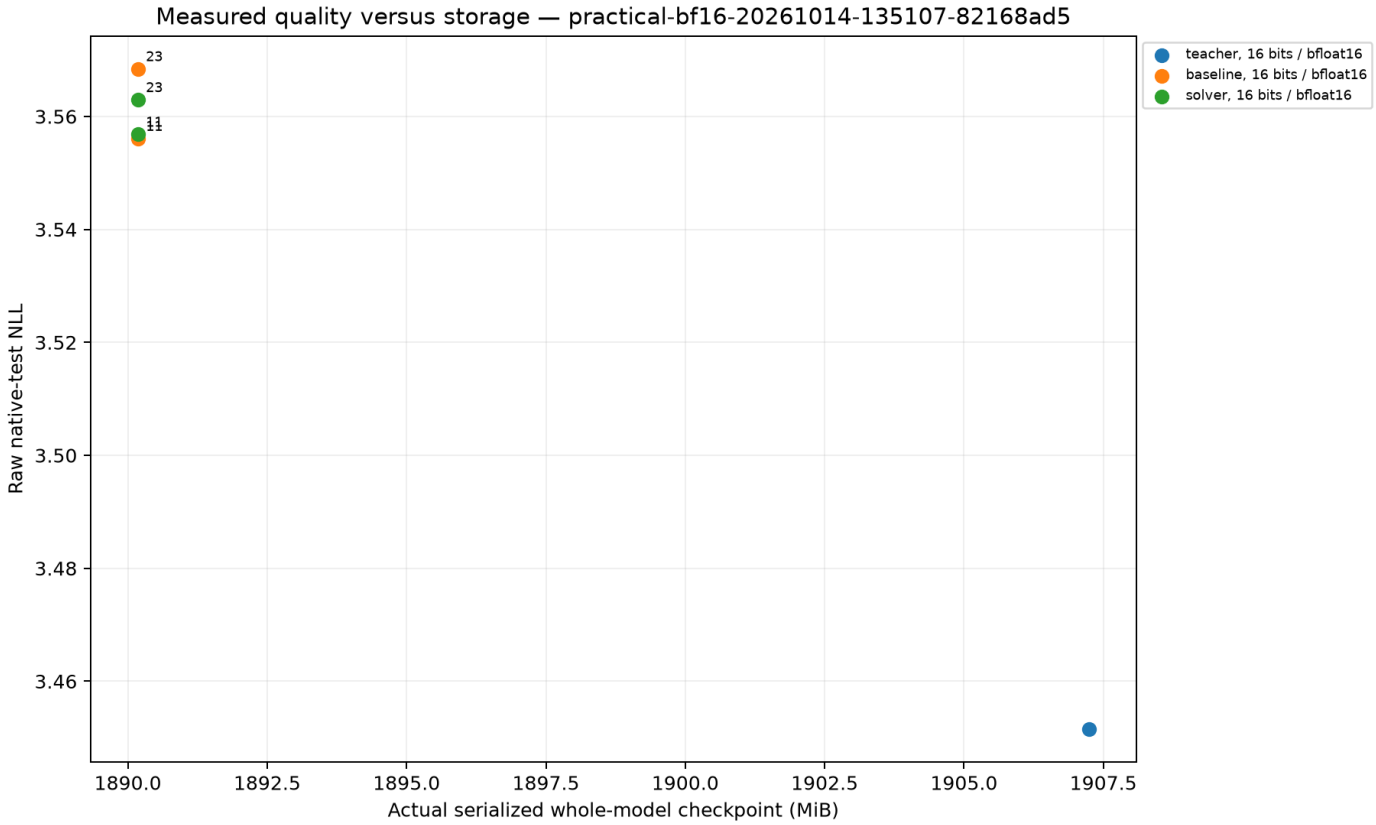
LiquidAI/LFM2.5-1.2B-Instruct: primary native-test teacher-relative NLL for completed exploratory cohort; individual paired seed points, mean and sample seed SD when at least two seeds exist. Actual target factor counts/fractions shown. Strict promotion remains disabled.



LiquidAI/LFM2.5-1.2B-Instruct: validation-only rank frontier for completed exploratory cohort. Each point aggregates only one exact rank recipe and available paired seeds; error bars are sample seed SD. X is actual retained target weight parameter fraction, not whole-model storage or a rank compression factor. Validation is not independent confirmation.



Gemma Stage C residual drift: individual seeds (dotted), mean (solid), sample seed SD (shaded). Single retained cohort; strict FP16 readiness remains failed.



Gemma Stage F raw native-test NLL versus actual whole-model checkpoint bytes; seed numbers label points. Factor-only full-rank controls excluded; no speed or lossless-compression claim.

Figure inputs, scope, source hashes and unavailable measurements are recorded in figures/provenance.json.

Review export formats

Standalone LaTeX source and a separately generated ReportLab PDF are review artifacts. Their preparation status and source hashes are recorded in export-status.json. The ReportLab PDF is not compiled from LaTeX; successful TeX compilation requires a separate compiler diagnostic.

Paired-block uncertainty appendix

Intervals are candidate minus the same-split baseline, using evaluation blocks rather than individual tokens. Partial-tail targets remain included in full NLL and are excluded from fixed-block intervals. These are not calibration-seed intervals.

architecture-replication-20261014-082807-5c73f669

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Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.24604289213573946	0.24615020975613225	[0.24006466116265526, 0.25237916880828776]	72/2000	2280

architecture-replication-20261014-082807-5c73f669

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 483, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 3, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.3.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 7, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.7.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 11, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.11.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 15, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.15.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 19, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.19.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}, {"block_index": 23, "dense_parameters": 2097152, "factor_parameters": 1483776, "name": "model.layers.23.self_attn.o_proj", "precision": "bfloat16", "rank": 483, "shape": [1024, 2048]}]}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.009941660681098283	0.010034525719555844	[0.00827384469021905, 0.01189700522595723]	72/2000	2280
23	0.008791898301914979	0.008937712051082021	[0.0072091521283551875, 0.010677830953140406]	72/2000	2280
37	0.007921984778739719	0.007972056491898503	[0.006199138578762092, 0.009875832550118127]	72/2000	2280

architecture-replication-20261014-082807-5c73f669

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Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.011256371692945919	0.011285927374891085	[0.00961544286242256, 0.013039805029695348]	63/2000	3362

architecture-replication-20261014-093845-97378b5e

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Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
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37	0.007921984778739719	0.007972056491898503	[0.006199138578762092, 0.009875832550118127]	72/2000	2280

architecture-replication-20261014-093845-97378b5e

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Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.011256371692945919	0.011285927374891085	[0.00961544286242256, 0.013039805029695348]	63/2000	3362
23	0.005501045557622271	0.005585379198455225	[0.0038466850466807364, 0.007429576815721795]	63/2000	3362
37	0.006284480060781636	0.006343413931392497	[0.004643602466799244, 0.008036439451197867]	63/2000	3362

architecture-replication-20261014-093845-97378b5e

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Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.027697128973382466	0.027779999025171056	[0.025335660561068965, 0.030283496110847567]	63/2000	3362
23	0.026820453601597936	0.027022706044184327	[0.024655860298807966, 0.02943987465715239]	63/2000	3362

architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "svd"}, "scope": {"rank": 725, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304,

"factor_parameters": 2969600, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}]}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.22646344793647272	-0.2264399873327979	[-0.24367673627667688, -0.20951701448693727]	73/2000	24

architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 725, "rank_divisor": null, "split": "test", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 2969600, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 725, "shape": [2048, 2048]}]}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.23260833182808122	-0.23258001440330633	[-0.24672731477834628, -0.21835409049405202]	73/2000	24
23	-0.20752383848706338	-0.20750846944987744	[-0.2240134011794247, -0.19023832839560784]	73/2000	24
37	-0.22133336282727098	-0.2213271375686348	[-0.23908078459653 28, -0.2034092545763355]	73/2000	24

architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 1024, "rank_divisor": 2, "split": "validation", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 4194304, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 1024, "shape": [2048, 2048]}]}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.1267830725542116	-0.1264940554545641	[-0.13380107717694656, -0.11900781495134909]	63/2000	3770
23	-0.11213960954857505	-0.1108899550184011	[-0.11815167612563775, -0.10308410012643247]	63/2000	3770
37	-0.12168323184273745	-0.12028804990045704	[-0.12827400810641157, -0.11168668095270724]	63/2000	3770

architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 682, "rank_divisor": 3, "split": "validation", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 2793472, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 682, "shape": [2048, 2048]}]}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.27847398152028663	-0.2766533448376106	[-0.2934387856600625, -0.25966835384794923]	63/2000	3770
23	-0.2610503911613548	-0.25815498372455165	[-0.2751110691947673, -0.24063632729316714]	63/2000	3770
37	-0.2795701052254451	-0.27724325879898043	[-0.29641486111395143, -0.2577876479611187]	63/2000	3770

architecture-replication-20261014-103759-8d32ab84

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"rank": 512, "rank_divisor": 4, "split": "validation", "target_recipes": [{"block_index": 2, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.2.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 5, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.5.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 8, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.8.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 10, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.10.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 12, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.12.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}, {"block_index": 14, "dense_parameters": 4194304, "factor_parameters": 2097152, "name": "model.layers.14.self_attn.out_proj", "precision": "bfloat16", "rank": 512, "shape": [2048, 2048]}]}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.34814383348227884	-0.3461968847176596	[-0.36750653404331235, -0.3257971909558875]	63/2000	3770
23	-0.29971982483469123	-0.2967208833552329	[-0.31580071090051415, -0.2776262464332278]	63/2000	3770
37	-0.3406797660259072	-0.3379485954229951	[-0.36191847249031006, -0.313945935288103]	63/2000	3770

baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "svd"}, "scope": {}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	1.5259270359038637	1.526963313415228	[1.4943535833878134, 1.5596674649359294]	71/2000	1465

baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"method": "asvd"}, "scope": {}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.13435744983933437	0.13448046045923576	[0.12208941117886445, 0.14684303527631282]	71/2000	1465
23	0.12406477628810314	0.12440374314135588	[0.1140738804768457, 0.13439828040507174]	71/2000	1465
37	0.10653445125745864	0.10667353359540374	[0.09858797073867313, 0.11492986771691276]	71/2000	1465

baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0031192516557841613	-0.003069222128221489	[-0.005598433325437209, -0.0005771062229211223]	71/2000	1465
23	-0.00867016563638367	-0.008560964606249673	[-0.01236161516313488, -0.004879390368432393]	71/2000	1465
37	-0.008888696495308254	-0.008889474302069626	[-0.011564302129920533, -0.006193025270745394]	71/2000	1465

baselines-20261013-175255-6a00d4e8

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.1, "method": "legacy_ridge", "ridge_fraction": 0.01}, "scope": {}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.007341743295384272	0.007358043732914079	[0.003578168563012017, 0.011322591469446464]	71/2000	1465
23	0.010085729923213327	0.010195171973247596	[0.005499654363179419, 0.01504279117529553]	71/2000	1465
37	0.0012338436455246004	0.0012706044011388724	[-0.002438464246056019, 0.00520269695332208]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-4.60899341388781e-05	-4.0872599009493115e-05	[-0.000217734233338580 93, 0.0001685818120544 4938]	71/2000	1465
23	-3.271375526325137e-05	-3.199588131774163e-05	[-0.000174187935932211 99, 0.0001136384687997568]	71/2000	1465
37	-6.647979225569334e-05	-7.095429546745745e-05	[-0.000291885207220884 4, 9.73636943746164e-05]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0008545763004039009	-0.0008561104717573092	[-0.001150257172183655 8, -0.00056394938574460 14]	71/2000	1465
23	-9.84808364817539e-05	-9.306537422449584e-05	[-0.000375805062010882 6, 0.00017279865033056 432]	71/2000	1465
37	-7.626569138096745e-05	-7.009637664775361e-05	[-0.000360900941821140 16, 0.00023415362379323]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.0001729304994845724 8	0.0001719993236095868 5	[4.463885584017398e-05, 0.0002936832496752188]	71/2000	1465
23	-0.0007079791885041549	-0.000699445138327318	[-0.000889761639764939 4, -0.000521697610992854]	71/2000	1465
37	-7.243953146688717e-05	-6.619463365792254e-05	[-0.000212401684606083 8, 7.996392602977384e-05]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0002764577707105269	-0.000305017713926657	[-0.000801674596175575 1, 0.00016777266329458 292]	71/2000	1465
23	0.0002748313234466563 5	0.0002560327026930727	[-0.000328662642760605 2, 0.0008294795855742868]	71/2000	1465
37	-0.0001416761346267915 2	-0.0001526383887316477 5	[-0.000643328925356078 7, 0.0003204156261228786]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12, 13], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0005983137392528166	-0.0005957751251080846	[-0.000967770908744049 3, -0.00022507084570211 577]	71/2000	1465
23	0.0004492915974461228	0.0004649281307566705 5	[5.583698743252717e-05, 0.0008733775102226093]	71/2000	1465
37	0.0001664124767705408 6	0.0001822159329953418 2	[-0.000260722413231894 8, 0.000609754279693661]	71/2000	1465

composition-20261013-185512-2bffd477

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [24, 25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	8.042555401250508e-05	8.104170434346619e-05	[-7.955390526647082e-05 , 0.000235949191167862 98]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-4.60899341388781e-05	-4.0872599009493115e-0 5	[-0.000217734233338580 93, 0.0001685818120544 4938]	71/2000	1465
23	-3.271375526325137e-05	-3.199588131774163e-05	[-0.000174187935932211 99, 0.0001136384687997568]	71/2000	1465
37	-6.647979225569334e-05	-7.095429546745745e-05	[-0.000291885207220884 4, 9.73636943746164e-05]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0008545763004039009	-0.0008561104717573092	[-0.001150257172183655 8, -0.00056394938574460 14]	71/2000	1465
23	-9.84808364817539e-05	-9.306537422449584e-05	[-0.000375805062010882 6, 0.00017279865033056 432]	71/2000	1465
37	-7.626569138096745e-05	-7.009637664775361e-05	[-0.000360900941821140 16, 0.00023415362379323]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.0001729304994845724 8	0.0001719993236095868 5	[4.463885584017398e-05, 0.0002936832496752188]	71/2000	1465
23	-0.0007079791885041549	-0.000699445138327318	[-0.000889761639764939 4, -0.000521697610992854]	71/2000	1465
37	-7.243953146688717e-05	-6.619463365792254e-05	[-0.000212401684606083 8, 7.996392602977384e-05]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0002764577707105269	-0.000305017713926657	[-0.000801674596175575 1, 0.00016777266329458292]	71/2000	1465
23	0.0002748313234466563 5	0.0002560327026930727	[-0.000328662642760605 2, 0.0008294795855742868]	71/2000	1465
37	-0.0001416761346267915 2	-0.0001526383887316477 5	[-0.000643328925356078 7, 0.0003204156261228786]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [12, 13], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0005983137392528166	-0.0005957751251080846	[-0.000967770908744049 3, -0.00022507084570211577]	71/2000	1465
23	0.0004492915974461228	0.0004649281307566705 5	[5.583698743252717e-05, 0.0008733775102226093]	71/2000	1465
37	0.0001664124767705408 6	0.0001822159329953418 2	[-0.000260722413231894 8, 0.000609754279693661]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [24, 25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	8.042555401250508e-05	8.104170434346619e-05	[-7.955390526647082e-05 , 0.000235949191167862 98]	71/2000	1465
23	-0.0008111692747406037	-0.0008061970833859053	[-0.001029544344414319 8, -0.00058981680664782 52]	71/2000	1465
37	0.0001807916412591724 4	0.0001831437652353506 8	[-9.780179775247661e-06 , 0.0003823935562813924]	71/2000	1465

composition-20261014-032120-486d799f

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1, 2, 3], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0006571169305328121	-0.0006716296099761055	[-0.00164636577805492, 0.0001938152765730989 4]	71/2000	1465
23	7.183583249946473e-05	6.899960868660811e-05	[-0.000682006490905095 6, 0.0007963317650939533]	71/2000	1465
37	-0.0002972410869361785 3	-0.0003128500272375454 3	[-0.000933935280597861 5, 0.0002562151766575395]	71/2000	1465

composition-20261014-032120-486d799f

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Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.003885308129297546	0.0039059187090884325	[0.0026608101493276353, 0.0051295292904874204]	71/2000	1465
23	0.003589996487403734	0.003661064528190402	[0.002338269696115027, 0.004953754396441149]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-4.60899341388781e-05	-4.0872599009493115e-05	[-0.0002177342333385809 3, 0.00016858181205444938]	71/2000	1465
23	-3.271375526325137e-05	-3.199588131774163e-05	[-0.0001741879359322119 9, 0.00011363846879975 68]	71/2000	1465
37	-6.647979225569334e-05	-7.095429546745745e-05	[-0.0002918852072208844, 9.73636943746164e-05]	71/2000	1465

composition-20261014-054423-dbd821ff

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Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0008545763004039009	-0.0008561104717573092	[-0.001150257172183655 8, -0.0005639493857446014]	71/2000	1465
23	-9.84808364817539e-05	-9.306537422449584e-05	[-0.0003758050620108826, 0.00017279865033056432]	71/2000	1465
37	-7.626569138096745e-05	-7.009637664775361e-05	[-0.0003609009418211401 6, 0.00023415362379323]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.0001729304994845724 8	0.0001719993236095868 5	[4.463885584017398e-05, 0.0002936832496752188]	71/2000	1465
23	-0.0007079791885041549	-0.000699445138327318	[-0.000889761639764939 4, -0.000521697610992854]	71/2000	1465
37	-7.243953146688717e-05	-6.619463365792254e-05	[-0.0002124016846060838, 7.996392602977384e-05]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0002764577707105269	-0.0003050177139266 57	[-0.000801674596175575 1, 0.00016777266329458292]	71/2000	1465
23	0.0002748313234466563 5	0.0002560327026930727	[-0.000328662642760605 2, 0.0008294795855742868]	71/2000	1465

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
37	-0.0001416761346267915 2	-0.0001526383887316477 5	[-0.000643328925356078 7, 0.0003204156261228786]	71/2000	1465

composition-20261014-054423-dbd821ff

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Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0005983137392528166	-0.0005957751251080846	[-0.000967770908744049 3, -0.00022507084570211 577]	71/2000	1465
23	0.0004492915974461228	0.0004649281307566705 5	[5.583698743252717e-05, 0.0008733775102226093]	71/2000	1465
37	0.0001664124767705408 6	0.0001822159329953418 2	[-0.000260722413231894 8, 0.000609754279693661]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [24, 25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	8.042555401250508e-05	8.104170434346619e-05	[-7.955390526647082e-05 , 0.000235949191167862 98]	71/2000	1465
23	-0.0008111692747406037	-0.0008061970833859053	[-0.001029544344414319 8, -0.00058981680664782 52]	71/2000	1465
37	0.0001807916412591724 4	0.0001831437652353506 8	[-9.780179775247661e-06 , 0.0003823935562813924]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1, 2, 3], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0006571169305328121	-0.0006716296099761055	[-0.00164636577805492, 0.0001938152765730989 4]	71/2000	1465
23	7.183583249946473e-05	6.899960868660811e-05	[-0.000682006490905095 6, 0.0007963317650939533]	71/2000	1465
37	-0.0002972410869361785 3	-0.0003128500272375454 3	[-0.000933935280597861 5, 0.0002562151766575395]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [11, 12, 13, 14], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	0.003885308129297546	0.0039059187090884325	[0.0026608101493276353 , 0.0051295292904874204]	71/2000	1465
23	0.003589996487403734	0.003661064528190402	[0.002338269696115027, 0.004953754396441149]	71/2000	1465
37	0.0006958575765264769	0.000706815205475545	[-0.000485747489089789 36, 0.0019332813322892252]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [22, 23, 24, 25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0004223128795064941	-0.00041862580332443627	[-0.000905342181576156, 8.412072718029896e-05]	71/2000	1465
23	-0.00290206343154077	-0.0029018436309528315	[-0.003464295129582087, -0.0023463511594901273]	71/2000	1465
37	0.0003427221577525009	0.0003274063402903456	[-0.000174796254365639, 0.000831103017138969]	71/2000	1465

composition-20261014-054423-dbd821ff

{"baseline": {"beta": 0.0, "method": "whitening", "ridge_fraction": 0.0}, "candidate": {"beta": 0.03, "method": "anchored", "ridge_fraction": 0.0}, "scope": {"group": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25], "rank": 384, "split": "test"}}

Seed	Full NLL delta	Subset delta	95% interval	Blocks/resamples	Excluded targets
11	-0.0031192516557841613	-0.003069222128221489	[-0.005598433325437209, -0.0005771062229211223]	71/2000	1465
23	-0.00867016563638367	-0.008560964606249673	[-0.01236161516313488, -0.004879390368432393]	71/2000	1465
37	-0.008888696495308254	-0.008889474302069626	[-0.011564302129920533, -0.006193025270745394]	71/2000	1465

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